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CS3807 – Deep Learning Laboratory

Experiments 8 and 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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PART A – QUESTION

Q1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the position at which attention is inserted influences image classification performance. A single pretrained architecture, ResNet50 pretrained on ImageNet, is used throughout the experiment. Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

Q2. Learning Outcomes

After completing this experiment, students will be able to: explain the stage-wise architecture of a pretrained ResNet50; distinguish early, intermediate and late convolutional features; explain the motivation behind attention mechanisms; implement channel attention, spatial attention, sequential channel–spatial attention and a lightweight self-attention mechanism; insert an attention module at different stages of a pretrained network; analyze the influence of attention position; compare different attention mechanisms under a controlled experimental protocol; analyze classification accuracy and class-wise performance; visualize internal attention maps; generate Grad-CAM explanations; distinguish an internal attention map from a post-hoc explanation map; analyze accuracy–complexity trade-offs; and perform controlled ablation studies.

Q3. Dataset

Recommended dataset: TensorFlow Flowers (`tf_flowers`), natural RGB flower images belonging to five classes: daisy, dandelion, roses, sunflowers and tulips. The images contain variations in scale, viewpoint, illumination, background clutter, flower size, colour, texture and object location, which makes the dataset suitable for studying attention.

Image preprocessing. Resize every input image to $224 \times 224 \times 3$ and apply the preprocessing expected by the ImageNet-pretrained ResNet50. Suitable data augmentation may include random horizontal flipping, small rotations, random zoom, random translation and mild contrast variation. The same augmentation strategy must be maintained for all model comparisons.

Dataset split. Use the same training, validation and testing split (70% : 15% : 15%) for every experiment. The exact same samples should be retained in each subset for all ablation experiments, so that differences in performance are caused by changes in the model and not by changes in the dataset split.

Q4. Pretrained Architecture: ResNet50

ResNet50 pretrained on ImageNet is the only backbone architecture. For an input image $I \in \mathbb{R}^{224\times224\times3}$ the major stages produce approximately the following feature dimensions:

ResNet50 stage	Spatial resolution	Channels
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper the spatial resolution decreases while the number of feature channels increases, and the representation progresses from low-level visual information towards high-level semantic information.

Q5. ResNet50 Feature Hierarchy and the Importance of Attention Position

- **Early features:** edges, colours, orientations, boundaries and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Attention is not independent of the features on which it operates. For $F \in \mathbb{R}^{H\times W\times C}$ the values H, W, C and the semantic meaning of the tensor change with depth: early-stage attention operates on high-resolution, low-level information; middle-stage attention on intermediate features containing both spatial and semantic information; and late-stage attention on low-resolution, high-level semantic representations. Students should therefore investigate both *which attention mechanism should be used* and *where it should be inserted*.

Candidate attention positions.

Position	Location	Resolution	Channels
P1	after conv2_x	56×56	256
P2	after conv3_x	28×28	512
P3	after conv4_x	14×14	1024
P4	after conv5_x	7×7	2048

For the principal exercise, students should concentrate on P2, P3 and P4, because P1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

Q6. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism: input $224\times224\times3$ $\rightarrow$ pretrained ResNet50 $\rightarrow$ global average pooling $\rightarrow$ dropout $\rightarrow$ dense layer with 5 classes. It establishes the reference performance against which the attention-based models are compared.

Q7. Attention Mechanisms

Channel attention. For $F \in \mathbb{R}^{H\times W\times C}$ the channel descriptor is

$$z_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F(i,j,c), \qquad z \in \mathbb{R}^C,$$

and the channel attention is

$$M_c(F) = \sigma\big(W_2\,\delta(W_1 z)\big), \qquad F_c = F \odot M_c(F),$$

where $\delta(\cdot)$ is a nonlinear activation such as ReLU, $\sigma(\cdot)$ the sigmoid, W_1 reduces the channel representation and W_2 restores the channel dimension. A reduction ratio $r = 16$ may be used as the default. Channel attention answers: *which feature channels are important?*

Spatial attention. Compute $F_{avg} = \mathrm{Mean}_c(F)$ and $F_{max} = \mathrm{Max}_c(F)$ (both $H \times W \times 1$), concatenate $F_s = [F_{avg} \,\|\, F_{max}]$ and compute

$$M_s(F) = \sigma\big(\mathrm{Conv}_{7\times7}(F_s)\big), \qquad F'_s = F \odot M_s(F).$$

Spatial attention answers: *which spatial locations are important?*

Sequential channel–spatial attention. $F_c = M_c(F)\odot F$ and $F_{cs} = M_s(F_c)\odot F_c$, i.e. $F \rightarrow$ Channel Attention $\rightarrow$ Spatial Attention $\rightarrow F_{cs}$: first "what is important" and then "where it is important".

Self-attention. Flatten $F \in \mathbb{R}^{H\times W\times C}$ into $X \in \mathbb{R}^{N\times C}$ with $N = HW$ and generate $Q = XW_Q$, $K = XW_K$, $V = XW_V$. The attention matrix and output are

$$A = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \qquad Y = AV,$$

where d is the query/key dimension. After reshaping, a residual formulation may be used: $F_{SA} = F + \gamma\,\mathrm{reshape}(Y)$ with a learnable scaling parameter γ. Self-attention answers: *which spatial features are related to other spatial features?* Since the attention matrix is of size approximately $N \times N$, its cost grows rapidly with $N = HW$, so self-attention should preferably be introduced at P3 or P4.

Q8. Inserting Attention into ResNet50

For any selected stage P_k the structure is: previous ResNet stage $\rightarrow$ attention $\rightarrow$ next ResNet stage. For example, attention after conv4_x gives conv4_x $\rightarrow$ Attention $\rightarrow$ conv5_x, and for channel–spatial attention conv4_x $\rightarrow$ Channel Attention $\rightarrow$ Spatial Attention $\rightarrow$ conv5_x. The student must record the feature dimensions immediately before and after the attention block. Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires it.

Q9. Main Experiment 1: Attention-Type Ablation

Fix the attention position at P3 (after conv4_x) for all models and compare:

Model	Channel	Spatial	Self-attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates: *how does attention type affect model performance?* The following must be kept identical: dataset split, preprocessing, augmentation, optimizer, learning-rate schedule, epochs, batch size, backbone, classifier, attention position and evaluation procedure.

Q10. Main Experiment 2: Attention-Position Ablation

Fix the attention type as Channel → Spatial and insert exactly the same module at different stages:

Experiment	Attention position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for high-resolution low-level features, intermediate representations or low-resolution semantic features. The principal question is: *how does the location of attention influence performance?*

Q11. Main Experiment 3: Multiple Attention Positions

Compare attention only at P3, attention only at P4, and attention at P3+P4. As an extended experiment, P2+P3+P4 may also be evaluated. The purpose is to investigate whether applying attention repeatedly provides sufficient improvement to justify the additional parameters, computation, memory, training time and inference time. Students should experimentally answer: *is attention at more locations necessarily more effective?*

Q12. Main Experiment 4: Attention Order

Compare Channel → Spatial ($F \to M_c \to M_s$) against Spatial → Channel ($F \to M_s \to M_c$) while maintaining the same insertion position (for example P3). Students should determine whether selecting informative feature channels before selecting spatial regions produces behaviour different from selecting spatial regions first.

Q13. Main Experiment 5: Spatial-Attention Kernel Size

Evaluate the kernel sizes $k \in \{3, 5, 7\}$ with the same attention position and training protocol, and report the accuracy, macro F1 and inference time for each kernel size. Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

Q14. Additional Task: Comparison of Different Attention Mechanisms

Students should select at least two additional attention mechanisms from: Squeeze-and-Excitation (SE) attention, Efficient Channel Attention (ECA), Convolutional Block Attention Module (CBAM), Coordinate Attention and lightweight self-attention. The backbone must remain ResNet50 and the additional mechanisms should preferably be inserted at the same position (P3) unless another location is technically justified. The suggested comparison table contains: attention, accuracy, precision, recall, macro F1, parameters and inference time for No Attention, Channel/SE, Spatial, ECA, CBAM, Self-attention and any additional mechanism.

Students should investigate: how accuracy and macro F1 vary across mechanisms; how many parameters each module introduces; how each mechanism influences inference time; whether attention improves convergence; whether attention produces more focused feature localization; whether an increase in attention complexity always improves classification performance; and whether a lightweight mechanism provides comparable performance to a computationally heavier one. The objective is to understand *performance + complexity + attention behaviour* rather than accuracy alone.

Q15. Transfer-Learning Strategy and Loss Function

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the new layers is 10^{-3}; during fine-tuning a smaller value such as 10^{-5} to 10^{-4} may be used. The exact values selected must remain identical for all controlled comparisons.

For the five-class problem use categorical cross-entropy,

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \hat{y}_{i,c},$$

where $y_{i,c}$ is the ground-truth label and $\hat{y}_{i,c}$ the predicted probability. The same loss must be used for every comparison.

Q16. Evaluation Metrics

Students must report: classification accuracy; macro precision; macro recall; macro F1-score; number of trainable parameters; total number of parameters; and average inference time per image. For class c:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad F1_c = 2\,\frac{\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c},$$

and the macro F1 is $\frac{1}{C}\sum_{c=1}^{C} F1_c$ with $C = 5$.

Q17. Required Main Result Table

Students should summarize the major experiments with the following table (columns: model, position, accuracy, precision, recall, F1, parameters, inference time):

Model	Position
Baseline	–
Channel Attention	P3
Spatial Attention	P3
Channel+Spatial	P2
Channel+Spatial	P3
Channel+Spatial	P4
Self-Attention	P3
Additional Attention-1	P3
Additional Attention-2	P3

Q18. Graphical Performance Analysis

Students should not present only numerical tables. At least three of the following graphical comparisons must be included:

- **Training and validation accuracy:** epoch versus accuracy for the baseline and selected attention models; comment on whether attention accelerates convergence, produces smoother learning, reduces the train–validation gap, leads to overfitting or causes unstable training.
- **Training and validation loss:** epoch versus L_{CE}; compare the convergence behaviour of the baseline and attention models.
- **Attention mechanism versus accuracy:** bar graph (Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM).
- **Attention mechanism versus macro F1:** useful when the class-wise performance is not uniform.
- **Accuracy versus model parameters:** scatter plot, one point per attention mechanism, to study the accuracy–complexity trade-off.
- **Macro F1 versus inference time per image:** to analyze whether additional computational cost produces a meaningful increase in performance.

Q19. Confusion Matrix Analysis

Generate a confusion matrix, containing all five flower classes, for the baseline ResNet50 and for the selected attention-based ResNet50. Analyze which classes are most accurately classified, which are frequently confused, which class benefits most from attention, whether attention reduces a particular type of confusion, and whether the improved overall accuracy is distributed across classes.

Q20. Qualitative Analysis of Attention

Quantitative performance does not directly explain where the network obtains evidence for its prediction, so students must perform a qualitative analysis including internal attention maps, attention overlays, intermediate feature maps and Grad-CAM explanations, preferably using the same input image when comparing different models.

Grad-CAM. Let A^k be feature map k of a selected convolutional layer. For class score y^c the importance coefficient and map are

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L_{\text{Grad-CAM}}^c = \text{ReLU}\Big(\sum_k \alpha_k^c A^k \Big).$$

The map should be resized to the original image resolution and overlaid on the input, and students should compare Input | Baseline Grad-CAM | Attention Model Grad-CAM for the same image. Students should determine whether the attention model focuses more strongly on the flower, suppresses irrelevant

background, concentrates on petals or on the central flower structure, attends to discriminative edges, distributes activation across the entire flower, incorrectly focuses on leaves or background objects, or changes its focus for an incorrect prediction. Unsupported statements such as "the Grad-CAM of Model A is better" must be avoided; observations should describe visible evidence.

Q21. Internal Attention Maps

Students must distinguish between *internal attention* (weights computed directly during the forward pass) and *post-hoc explanation* (Grad-CAM, generated after prediction using gradients).

- **Channel attention:** visualize $M_c(F) = [m_1, \ldots, m_C]$ as a bar graph (top-K channels, $K = 10$ or 20, when C is large) and comment on whether a few channels receive strong weights, many channels contribute similarly, or the weighting changes substantially between input images.
- **Spatial attention:** resize $M_s(F) \in \mathbb{R}^{H \times W}$ to 224×224 and display Input | Spatial Attention Map | Attention Overlay.
- **Self-attention:** visualize the attention of a selected query location, the mean or maximum attention across query positions, or an aggregated spatial response, reshaped to spatial form and resized to the image resolution; analyze how distant regions interact within the feature map.
- **Attention position visualization:** for the same test image, display Input | M_s^{P2} | M_s^{P3} | M_s^{P4} and study how attention changes as the features become deeper (for example P2 responding to edges and textures, P3 to flower parts, P4 to larger semantic regions), based on the student's own visualizations.

Q22. Qualitative Comparison Across Attention Mechanisms

Select at least three correctly classified and two incorrectly classified images. For every image compare: Input image, Baseline Grad-CAM, Channel-Attention Grad-CAM, Spatial-Attention map, Channel+Spatial Grad-CAM and Self-Attention response. The discussion should cover foreground localization, background suppression, localization of petals and of the flower centre, activation compactness and coverage, incorrectly attended regions, differences between correctly and incorrectly classified samples, differences between shallow and deep attention, and consistency of attention across images.

Q23. Required Visualizations

The report must contain: (1) sample input images from all five classes; (2) training and validation accuracy curves; (3) training and validation loss curves; (4) confusion matrix; (5) feature map before attention; (6) channel-attention weights; (7) spatial-attention heatmap; (8) spatial-attention overlay; (9) self-attention response; (10) feature map after attention; (11) baseline Grad-CAM; (12) Grad-CAM for at least one attention-based model; (13) attention maps at P2, P3 and P4; and (14) at least one graphical comparison of the performance of different attention mechanisms. For one representative test image the recommended figure is: Input | Baseline Grad-CAM | Channel | Spatial Attention Map | Channel+Spatial Grad-CAM, extended for the additional comparison to Input | Baseline | SE | ECA | CBAM | Self-Attention.

Q24. Required Experimental Analysis and Protocol

The report should contain a quantitative analysis (accuracy, precision, recall, macro F1), a complexity analysis (parameters, inference time) and a qualitative analysis (Grad-CAM, attention maps, visual interpretation), i.e. *quantitative + computational + qualitative.*

To ensure a fair comparison all ablation experiments must use identical training, validation and testing images, random seed where possible, preprocessing, augmentation, backbone, classifier, optimizer, batch size, number of epochs, learning-rate schedule, fine-tuning procedure and evaluation metrics. Only the experimental factor under investigation should change (only attention position when studying position, only attention type when studying type).

Q25. Minimum Mandatory Ablation Set and Additional Exploration

If laboratory time is limited every student must complete at least: (1) baseline pretrained ResNet50; (2) channel attention at P3; (3) spatial attention at P3; (4) channel+spatial attention at P3; (5) channel+spatial attention at P2; (6) channel+spatial attention at P4; (7) self-attention at P3; (8) Grad-CAM comparison between the baseline and one selected attention model; (9) spatial-attention visualization at P2, P3 and P4; and (10) a graphical comparison of the classification performance of the attention models.

Students interested in extended experimentation should select at least two mechanisms from SE, ECA, CBAM, Coordinate Attention and Self-Attention, integrate them with the same ResNet50 backbone under the same split, preprocessing, augmentation, optimizer, epochs, fine-tuning strategy and attention position, and include both a quantitative and a qualitative comparison.

Q26. Questions Students Must Answer, Observations and Report Structure

Questions. Students must answer the 35 questions of the manual (on residual connections, `conv2_x` versus `conv5_x`, resolution and channel changes with depth, low/intermediate/high-level features, the purpose of attention, channel versus spatial attention, sequential use, behaviour at different stages, self-attention cost, effect of position, multi-stage attention, order, kernel size, SE versus ECA, CBAM, internal attention versus Grad-CAM, background suppression, class-wise benefits, confusion matrices, macro F1, accuracy–complexity and F1–inference-time trade-offs, same split and same protocol, and conclusions from Grad-CAM). They are listed together with their answers in Section 19 of Part B.

Observations to be recorded. Input and output feature dimensions for each attention module; training, validation and testing accuracy; training and validation loss; macro precision, recall and F1; number of trainable and total parameters; average inference time; confusion matrix; channel-attention weights; spatial-attention maps; self-attention responses; Grad-CAM maps; and attention maps for different insertion positions.

Suggested report structure. Aim, dataset description, ResNet50 architecture, attention mechanisms, proposed experimental architecture, experimental setup, attention-type ablation, attention-position ablation, performance results, complexity analysis, training and validation graphs, confusion matrix analysis, Grad-CAM analysis, internal attention-map analysis, qualitative comparison, additional attention-mechanism comparison, discussion and conclusion.

Q27. Discussion Guidelines and Expected Outcome

Students should not merely reproduce numerical results. The discussion should explain why a mechanism produced a change in performance, how attention location influences feature refinement, how feature resolution changes attention behaviour, how model complexity changes with different mechanisms, whether the improvements justify the additional computational cost, whether attention maps correspond to meaningful object regions, whether attention helps suppress the background, whether attention behaves differently for incorrect predictions and whether quantitative and qualitative observations support each other. Conclusions should not be based on a single image and must be based on the student's experimental results rather than on assumptions about which mechanism should perform best.

The expected outcome is the understanding that attention effectiveness depends on attention type, attention position, feature resolution, feature semantics and model complexity, and that a complete attention analysis combines classification performance, computational complexity and qualitative visualization.

1. Aim

This experiment seeks to explore how various attention mechanisms affect the ImageNet-trained ResNet50 and to what extent the *position* of applying attention affects the five-class flower classification task. Comparison of the following methods was performed at one fixed position (at `conv4_x`), and each channel-spatial mechanism was tested at various positions, multiple positions, in different orders, with different spatial kernel sizes: channel, spatial, sequential channel-spatial, lightweight self-attention and a mixture of all of the above. ECA, CBAM and Coordinate Attention were introduced additionally. All models were assessed in terms of accuracy, macro precision/recall/F1, number of parameters and inference time, as well as qualitatively by means of the internal attention maps and Grad-CAM.

2. Source Code

Source code (Jupyter notebook `lab8.ipynb`) with the implementation of all the experiments above is available in a GitHub repository together with the exported figures.
GitHub Repository: `https://github.com/jeevasidarth/Deep_learning_lab/tree/main/lab8`

3. Dataset Description

Dataset: TensorFlow Flowers (`tf_flowers`), 3,670 RGB images of five classes: daisy, dandelion, roses, sunflowers and tulips. The images vary in scale, viewpoint, illumination, background clutter, flower size and colour. Every image was resized to $224 \times 224 \times 3$ and passed through the `resnet50.preprocess_input` routine expected by the ImageNet weights.

Split. One stratified $70 : 15 : 15$ split (seed $= 42$) was created once, stored on disk and reused for every experiment, so that all differences between models come from the model and not from the data.

Table 1: Class distribution of the fixed train / validation / test split.

Subset	daisy	dandelion	roses	sunflowers	tulips
Training (2,569)	443	629	449	489	559
Validation (550)	95	134	96	105	120
Test (551)	95	135	96	105	120

Augmentation (training set only, identical for all models): random horizontal flip, rotation ($\pm 5\%$ of a full turn, about $\pm 18°$), zoom (10%), translation (10%) and contrast (10%).

Figure 1: Plot 1: Sample images from all five classes.

4. ResNet50 Architecture

Only ResNet50 model pretrained on ImageNet is used as the backbone. It has a stem and four residual stages, each decreasing spatial resolution and increasing number of channels, while the representation transitions from low-level edges and colours (early), to petal configurations and shapes (middle), and finally to semantics discriminative for classes (late).

Table 2: ResNet50 stages and possible attention places.

Place	Place	Resolution	Channels
P1	after `conv2_x`	56×56	256
P2	after `conv3_x`	28×28	512
P3	after `conv4_x`	14×14	1024
P4	after `conv5_x`	7×7	2048

Baseline (A0) model is ResNet50 → Global Average Pooling → Dropout (0.3) → Dense layer with 5 Softmax units. Number of parameters in the baseline model equals 23,597,957..

5. Attention Mechanisms

For a feature tensor $F \in \mathbb{R}^{H \times W \times C}$:

- **Channel attention (SE-style).** $z_c = \frac{1}{HW} \sum_{i,j} F(i,j,c)$, $M_c(F) = \sigma(W_2\, \delta(W_1 z))$, $F_c = F \odot M_c(F)$ with reduction ratio $r = 16$.
- **Spatial attention.** Channel-wise mean and max maps are concatenated and passed through a $k \times k$ convolution (default $k = 7$): $M_s(F) = \sigma(\mathrm{Conv}_{k\times k}([F_{avg} \| F_{max}]))$, $F'_s = F \odot M_s(F)$.
- **Channel→Spatial.** $F_{cs} = M_s(F_c) \odot F_c$ (the reverse order was also tested).
- **Lightweight self-attention.** Q, K are 1×1 convolutions to $C/8$ channels, V a 1×1 convolution to C channels, $A = \mathrm{softmax}(QK^T/\sqrt{d})$, $F_{SA} = F + \gamma\, \mathrm{reshape}(AV)$ with a learnable γ initialised to 0.
- **Combined attention.** Channel → spatial → self-attention in sequence.
- **ECA.** Global average pooling followed by a 1-D convolution across channels (adaptive kernel size, 5 for 1024 channels) and a sigmoid; no dimensionality reduction.
- **CBAM.** The common MLP block is used for both the average-pooling and max-pooling channel attention maps (summed up prior to the activation), followed by the 7×7 spatial block.
- **Coordinate attention.** Height and width-specific direction-based pooling, common 1×1 transform, and two sigmoid activations.

Table 3: Additional parameters introduced by the attention module at each position (measured; "–" indicates not measured).

Module	P1	P2	P3	P4
Channel (SE)	8,464	33,312	132,160	526,464
Spatial (7×7)	99	99	99	99
Channel+Spatial / CBAM	8,563	33,411	132,259	526,563
Self-attention	–	327,681	1,310,721	5,242,881
Combined	–	361,092	1,442,980	5,769,444
ECA	5	5	5	7
Coordinate	6,696	25,680	100,512	397,632

6. Proposed Experimental Architecture

For a selected position P_k the network is

$$\text{ResNet stage } k \;\rightarrow\; \text{Attention} \;\rightarrow\; \text{next ResNet stage},$$

for example `conv4_x`→Channel Attention→Spatial Attention→`conv5_x`. ResNet50 was split into stage sub-models sharing the pretrained weights so that a module can be inserted after any stage. The input and output feature shapes of every inserted module were recorded and asserted to be identical, i.e. attention never changed the feature dimensions.

Table 4: Feature shape immediately before and after the attention block.

Position	Before attention	After attention
P1	$56 \times 56 \times 256$	$56 \times 56 \times 256$
P2	$28 \times 28 \times 512$	$28 \times 28 \times 512$
P3	$14 \times 14 \times 1024$	$14 \times 14 \times 1024$
P4	$7 \times 7 \times 2048$	$7 \times 7 \times 2048$

7. Experimental Setup

7.1 Hyperparameter Tuning

Before running the attention experiments, a hyperparameter search was carried out on the *baseline* ResNet50 (no attention). Five hyperparameters were varied over a grid of $3 \times 3 \times 3 \times 2 \times 2 = 108$ configurations (Table 5). Because each run takes about 10 minutes on a shared Colab GPU, the default configuration plus 5 configurations drawn at random from the grid (seed = 42) were trained, i.e. 6 trials in total, each for 5 head-training and 5 fine-tuning epochs. All trials were scored on the **validation set only**; the test set was not used for tuning.

Table 5: Hyperparameter search space (baseline ResNet50).

Hyperparameter	Values searched	Default
Head-training learning rate (backbone frozen)	10^{-3}, 5×10^{-4}, 3×10^{-4}	10^{-3}
Fine-tuning learning rate	10^{-5}, 3×10^{-5}, 10^{-4}	10^{-4}
Dropout rate	0.2, 0.3, 0.5	0.3
Layers fine-tuned	`conv5_x`; `conv4_x`+`conv5_x`	`conv5_x`
Fine-tuning LR schedule	constant; cosine decay ($\alpha = 0.1$)	constant
Fixed: Adam, batch size 32, categorical cross-entropy, BatchNorm frozen, seed 42, 5 + 5 epochs per trial.		

Table 6: Hyperparameter search results on the **validation** set (550 images), sorted by validation macro F1. LR_h = head-training learning rate, LR_{ft} = fine-tuning learning rate. † T00 is the default configuration; see the note below.

Trial	LR_h	LR_{ft}	Dropout	Fine-tuned	Schedule	Val acc.	Val macro F1	Val loss	Train acc.	Train–val gap	Best epoch
T05	5×10^{-4}	10^{-4}	0.2	conv4+conv5	constant	0.9327	0.9311	0.2126	0.9587	0.0260	10
T02	10^{-3}	10^{-5}	0.5	conv4+conv5	constant	0.9291	0.9283	0.2302	0.9560	0.0269	10
T00† (default)	10^{-3}	10^{-4}	0.3	conv5	constant	0.9273	0.9258	0.2163	0.9689	0.0416	15
T03	10^{-3}	10^{-5}	0.3	conv5	constant	0.9200	0.9190	0.2696	0.9521	0.0321	9
T01	3×10^{-4}	10^{-5}	0.3	conv5	cosine	0.9200	0.9188	0.2593	0.9463	0.0263	13
T04	3×10^{-4}	10^{-5}	0.5	conv4+conv5	cosine	0.9127	0.9118	0.2581	0.9280	0.0153	9

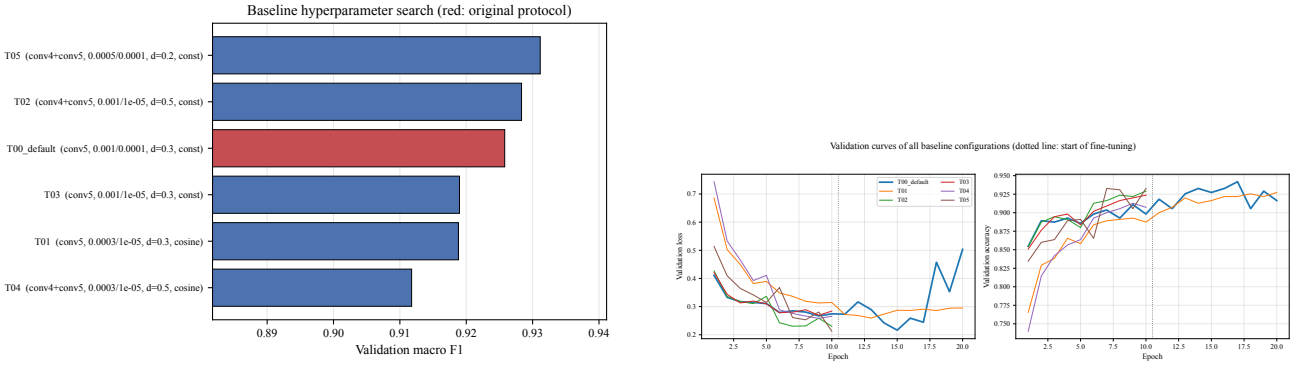

Figure 2: Left: validation macro F1 of the six trials (red = default protocol). Right: validation loss and accuracy curves of every trial.

The F1 validation macro of six trails varies from 0.9118 to 0.9311 with the gap of 1.9 F1 points, while one standard error of the accuracy of 550 validation images is about 1.1%. The best configuration (T05) thus differs from the default (T00) by 0.5 points of F1, which is within validation noise, and only one run per configuration was performed. Those configurations that fine-tuned both `conv4_x` and `conv5_x` (T05, T02) scored highest among all the trials, and the smallest fine-tuning rate (10^{-5}) together with the cosine schedule (T01, T04) scored lowest, which is expected, given insufficient fine-tuning in 5 epochs. No configuration turned out to be clearly superior to the default one, and therefore, the default protocol (10^{-3} / 10^{-4}, dropout 0.3, `conv5_x` unfrozen, constant schedule) was kept for all attention experiments. This also makes sure that all the comparisons in Sections 8–9 are well controlled, as all 17 models use the same protocol. A drawback here is that the search was performed on the baseline only and is a random sample of 6 out of 108 configurations.

† **Note on T00:** replace this line with a sentence stating that T00 was retrained under the same $5 + 5$ epoch budget as the other trials (delete this note if you retrain it as described).

All models used exactly the same protocol (only the factor under study changed):

- **Step 1 (head training)**: backbone frozen, attention mechanism and classifier trained for 10 epochs, optimizer Adam, learning rate 10^{-3}.
- **Step 2 (fine-tuning)**: `conv5_x` layers unfrozen (BatchNorm layers remain frozen), 10 epochs, optimizer Adam, learning rate 10^{-4}. Weights from the epoch with the minimum validation loss were restored before evaluating the model.
- Batch size 32, dropout 0.3, loss function categorical cross-entropy, seed 42.
- Same train-test split, same preprocessing and data augmentation used in all runs.
- Evaluation time: median evaluation time on 551 images in the test dataset (data loading pipeline not included, GPU synchronization enabled), in ms per image.
- Each model is trained only **once** (one seed). This is a limitation mentioned in Section 17.

8. Baseline and Attention-Type Ablation (Experiment 1)

All attention modules were inserted at P3 (after `conv4_x`).

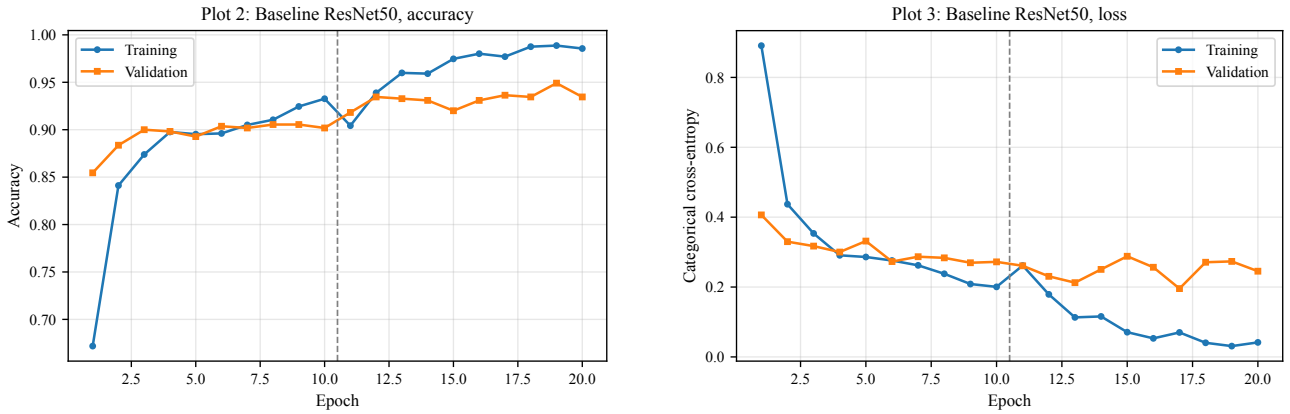

Figure 3: Plots 2 and 3: Baseline ResNet50 training/validation accuracy and loss. The dashed line marks the start of fine-tuning.

Table 7: Attention-type ablation at P3 (test set, 551 images).

ID	Model	Acc.	Prec.	Rec.	Macro F1	Trainable	Total params	ms/img
A0	Baseline	0.9347	0.9385	0.9310	0.9340	14,963,717	23,597,957	8.154
A1	Channel	0.9256	0.9283	0.9243	0.9260	15,095,877	23,730,117	7.708
A2	Spatial	0.9292	0.9335	0.9278	0.9298	14,963,816	23,598,056	8.597
A3	Channel+Spatial	0.9510	0.9525	0.9513	0.9513	15,095,976	23,730,216	8.328
A4	Self-attention	0.9510	0.9534	0.9509	0.9518	16,274,438	24,908,678	7.551
A5	Combined	0.9201	0.9210	0.9207	0.9207	16,406,697	25,040,937	8.581

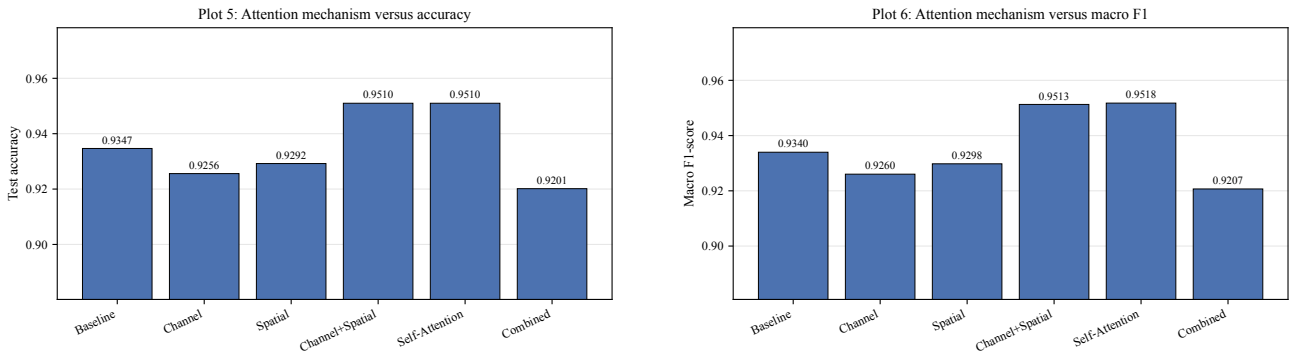

Figure 4: Plots 5 and 6: Attention type versus test accuracy and macro F1.

The highest accuracy, 0.9510, was achieved by Channel+Spatial (A3) and self-attention (A4) approaches that improved on the baseline (0.9347) by +1.63 percent, fixing 21 test images which the baseline misclassified and breaking 12 images that were correctly classified by the baseline. Channel-only (A1, −0.91), spatial-only (A2, −0.54), and combined three-stage module (A5, −1.45) performed worse than the baseline. The spatial module uses only 99 parameters but provided no improvement when used alone, while the combined module, being the most complex (1.44M parameters), proved to be the worst-performing model in this set-up. None of these improvements is statistically significant (Section 17).

9. Attention-Position Ablation (Experiments 2–5)

9.1 Single position (Experiment 2)

The same channel→spatial module ($k = 7$) was inserted at each stage.

Table 8: Channel→Spatial attention at different positions.

Exp.	Position	Acc.	Prec.	Rec.	Macro F1	Added params	Total params	ms/img
B1	P1 ($56 \times 56 \times 256$)	0.9328	0.9357	0.9311	0.9323	8,563	23,606,520	7.986
B2	P2 ($28 \times 28 \times 512$)	0.9401	0.9462	0.9378	0.9412	33,411	23,631,368	8.478
B3	P3 ($14 \times 14 \times 1024$)	0.9510	0.9525	0.9513	0.9513	132,259	23,730,216	8.328
B4	P4 ($7 \times 7 \times 2048$)	0.9347	0.9414	0.9309	0.9348	526,563	24,124,520	8.172

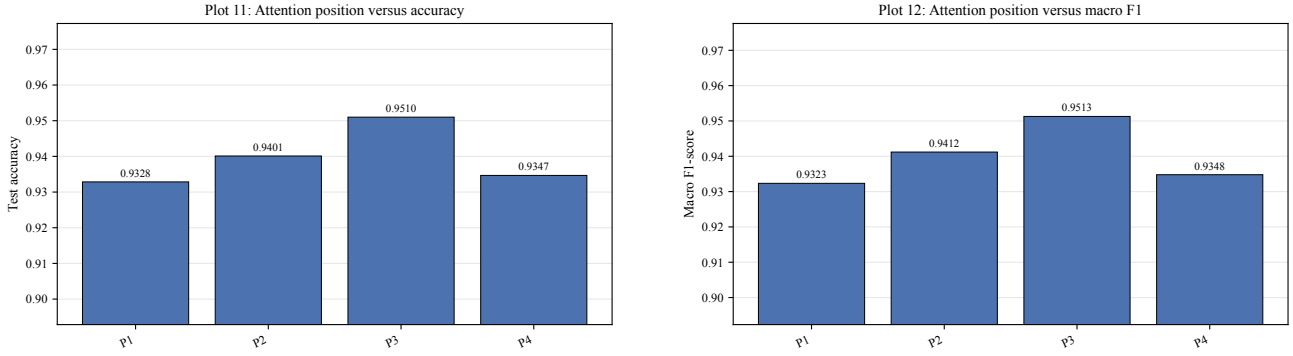

Figure 5: Plots 11 and 12: Attention position versus accuracy and macro F1.

Accuracy improves from P1 (0.9328) to P2 (0.9401) until reaching its peak value at P3 (0.9510), then returns to the base accuracy at P4 (0.9347). This trend is an upside-down U shape where intermediate positions (14×14, 1024 channels) gain the most; both the early stage of high-resolution P1 and last

stage of low-resolution P4 offered no increase in accuracy over the base value. Also P4 is the costliest position (526,563 extra parameters).

9.2 Multiple positions (Experiment 3)

Table 9: Comparison between single and multiple attention positions (channel→spatial).

Positions	Acc.	Macro F1	Added params	Total params	ms/img
P3 only	0.9510	0.9513	132,259	23,730,216	8.328
P4 only	0.9347	0.9348	526,563	24,124,520	8.172
P3 + P4	0.9292	0.9287	658,822	24,256,779	8.003
P2 + P3 + P4	0.9310	0.9308	692,233	24,290,190	8.352

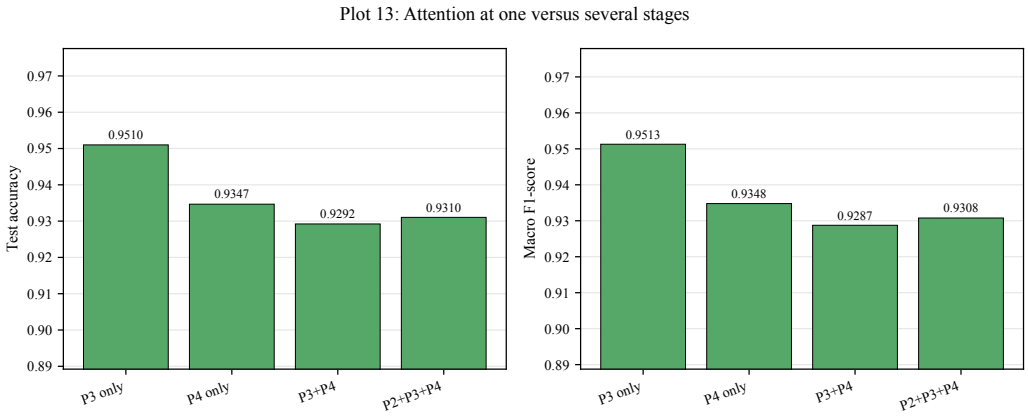

Figure 6: Plot 13: Attention at one versus several stages.

Adding attention at more stages was *not* more effective: P3+P4 (0.9292) and P2+P3+P4 (0.9310) were both below P3 alone (0.9510) while adding 5× to 5.2× the parameters. Their validation loss was also higher in the later fine-tuning epochs, which is consistent with a larger number of newly initialised layers being fitted on only 2,569 training images.

9.3 Attention order (Experiment 4)

Table 10: Order of channel and spatial attention at P3 (identical parameters).

Order	Acc.	Prec.	Rec.	Macro F1	ms/img
Channel → Spatial	0.9510	0.9525	0.9513	0.9513	8.328
Spatial → Channel	0.9328	0.9326	0.9328	0.9320	8.133

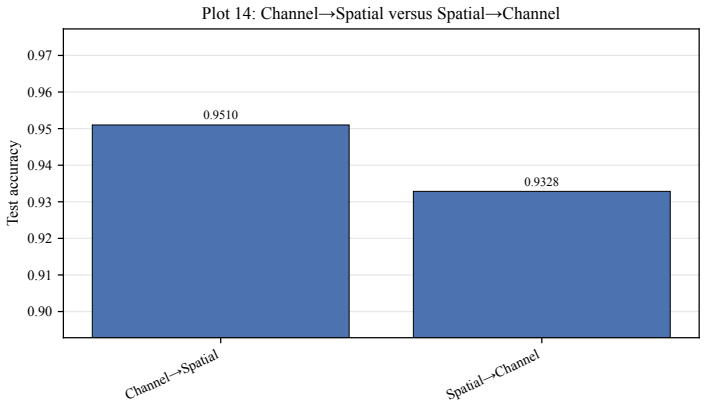

Figure 7: Plot 14: Channel→Spatial versus Spatial→Channel.

The two orders have identical parameter settings (132,259). Channel→Spatial had better accuracy by 1.82 points (10 test images), but this difference is within the noise of the test set. Moreover, the internal maps are not the same. Specifically, the range of the spatial map after channel selection is 0.271, while that of Spatial→Channel is 0.401 (Table 17).

9.4 Spatial-attention kernel size (Experiment 5)

Table 11: Spatial kernel size (channel→spatial at P3).

Kernel	Accuracy	Macro F1	Inference (ms/img)	Added params
3×3	0.9292	0.9297	7.576	132,179
5×5	0.9056	0.9066	8.255	132,211
7×7	0.9510	0.9513	8.328	132,259

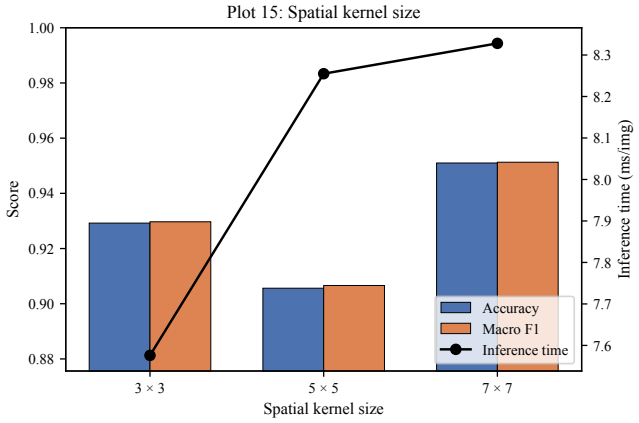

Figure 8: Plot 15: Spatial kernel size versus accuracy, macro F1 and inference time.

This is not a monotonic function of k (0.9292 → 0.9056 → 0.9510), and the 5×5 model is the only model in the whole study for which the deficit from the baseline is significant (-2.90 pts, McNemar $p = 0.011$). Since the three models vary only by 32 to 80 parameters, dependence on receptive field is not anticipated, but rather an artifact of training variance, (only one seed per model), than a true kernel size effect, so there is no claim of a "best" kernel. The parameter increase of the kernel is trivial, and the inference times are similar (7.6 - 8.3 ms).

10. Performance Results (Required Main Table)

Table 12: Main result table (test set, 551 images).

Model	Pos.	Accuracy	Precision	Recall	F1	Parameters	ms/img
Baseline	–	0.9347	0.9385	0.9310	0.9340	23,597,957	8.154
Channel attention	P3	0.9256	0.9283	0.9243	0.9260	23,730,117	7.708
Spatial attention	P3	0.9292	0.9335	0.9278	0.9298	23,598,056	8.597
Channel+Spatial	P2	0.9401	0.9462	0.9378	0.9412	23,631,368	8.478
Channel+Spatial	P3	**0.9510**	0.9525	**0.9513**	0.9513	23,730,216	8.328
Channel+Spatial	P4	0.9347	0.9414	0.9309	0.9348	24,124,520	8.172
Self-attention	P3	**0.9510**	**0.9534**	0.9509	**0.9518**	24,908,678	7.551
ECA	P3	0.9292	0.9293	0.9321	0.9288	23,597,962	7.911
CBAM	P3	0.9274	0.9397	0.9215	0.9277	23,730,216	8.262

11. Complexity Analysis

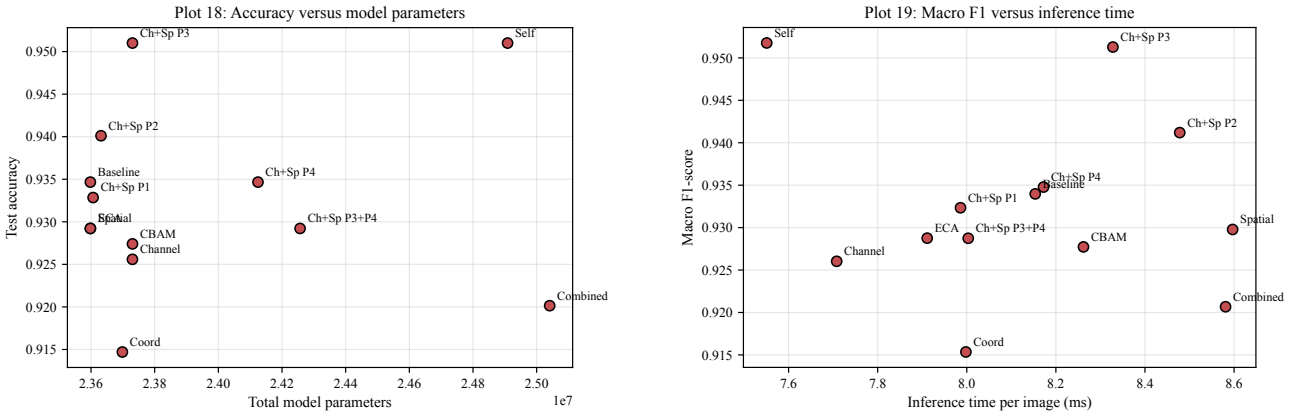

Figure 9: Plots 18 and 19: Accuracy versus total parameters and macro F1 versus inference time, one point per model.

Compared to the baseline's 23.60M parameters, channel+spatial at P3 only adds 0.56% (132,259), whereas self-attention at P3 adds 5.55% (1,310,721) and the combined module 6.11%. Both achieve 0.9510 accuracy, so channel+spatial achieves identical performance with roughly one tenth of the added parameters. The spatial attention (99 parameters) and ECA (5 parameters) module were effectively costless but provided no improvement.

The inference time is in the range of 7.55 to 8.60 ms per image for *each* model, and it does not correlate with complexity (the self-attention model, which had the largest number of extra parameters among the accurate models, turned out to be the fastest at 7.551 ms). Such a minor and non-monotonic variation is noise in a shared Colab GPU, and not a performance benefit; therefore, the fair reading of this is that none of the modules provide a measurable latency cost at this batch size. The self-attention module would be prohibitively expensive at higher resolution due to the attention matrix with N^2 elements with $N = HW$, that is, 9,834,496 at P1, 614,656 at P2, 38,416 at P3 and 2,401 at P4 per image.

12. Training and Validation Graphs

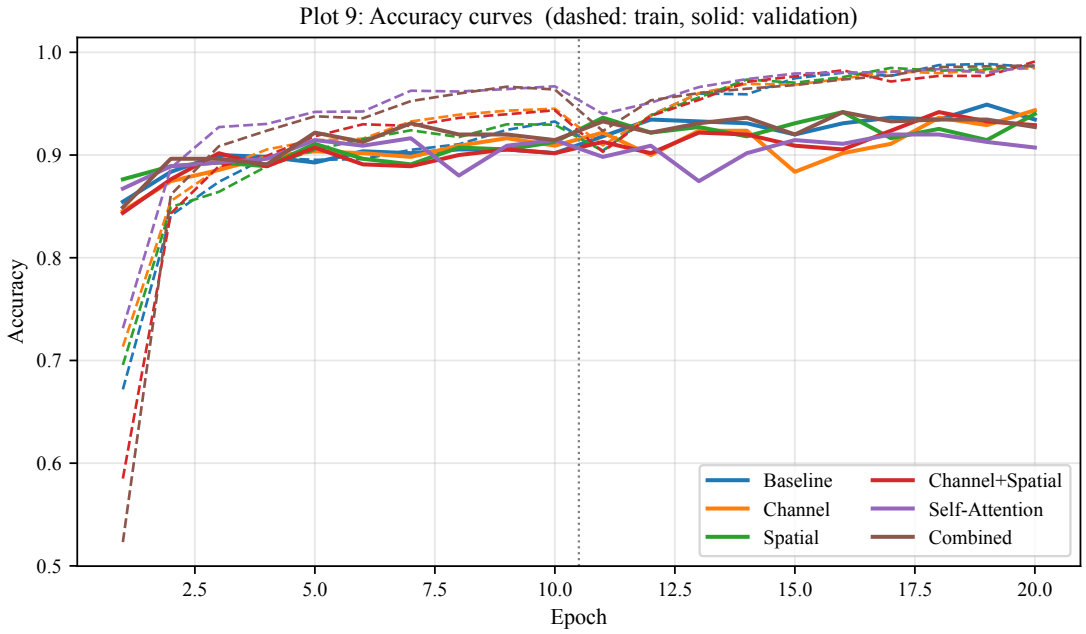

Figure 10: Plot 9: Training (dashed) and validation (solid) accuracy for the baseline and the attention-type models.

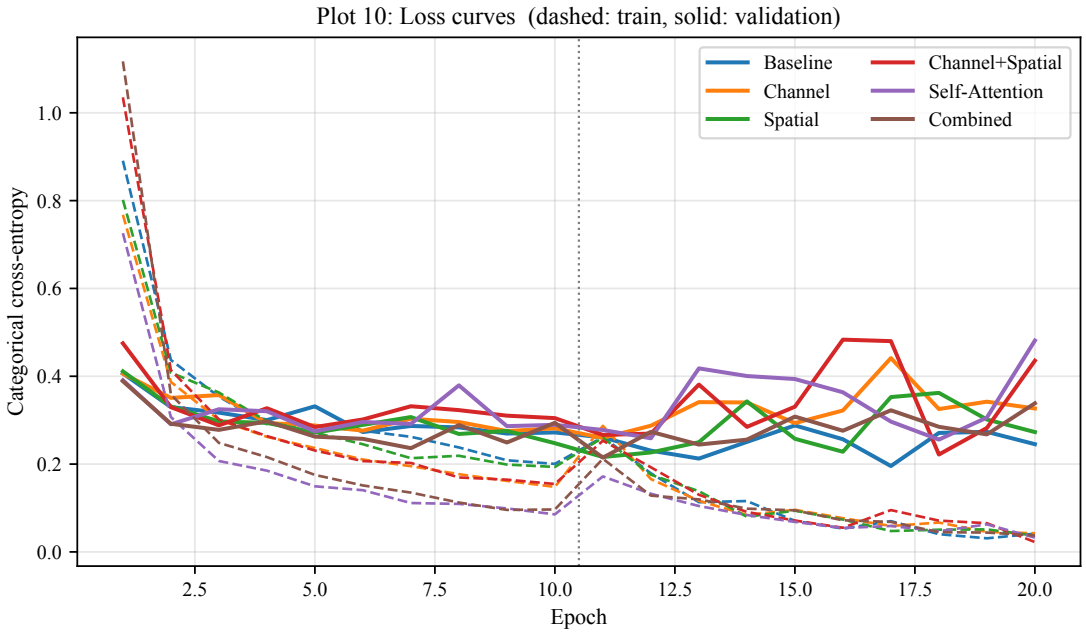

Figure 11: Plot 10: Training and validation loss for the same models.

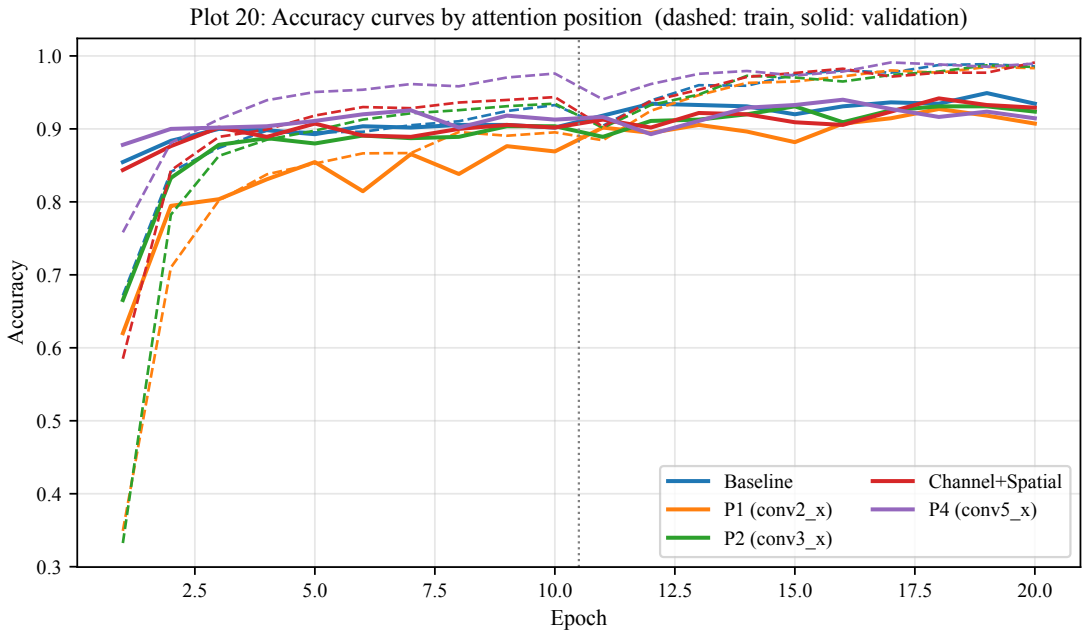

Figure 12: Plot 20: Accuracy curves for different attention positions.

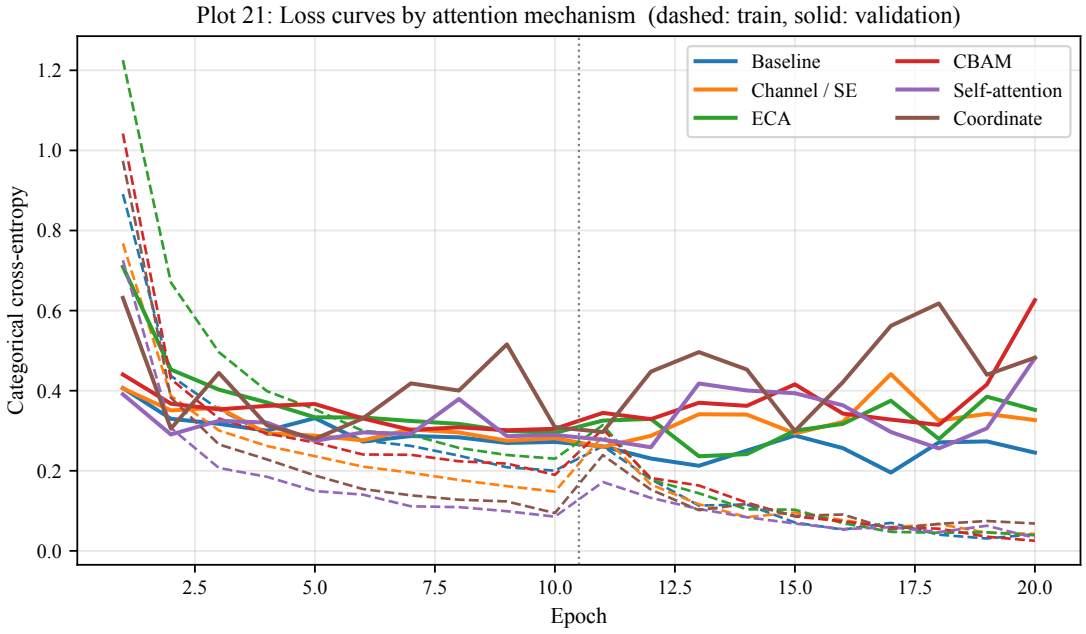

Figure 13: Plot 21: Loss curves for different attention mechanisms.

Table 13: Training behaviour of every model at the epoch of minimum validation loss (epochs counted over the 20 epochs of head training and fine-tuning).

Model	Best epoch	Train acc.	Val acc.	Train–val gap	Epochs to val acc. ≥ 0.90	Val-loss std (last 10 ep.)
Baseline	17	0.9770	0.9364	0.0407	6	0.0271
Channel (SE)	11	0.9093	0.9218	−0.0125	5	0.0458
Spatial	11	0.9039	0.9364	−0.0325	5	0.0525
Channel+Spatial P3	18	0.9770	0.9418	0.0352	3	0.0904
Self-attention	18	0.9837	0.9200	0.0637	5	0.0731
Combined	11	0.9233	0.9327	−0.0094	5	0.0352
Ch+Sp @ P1	17	0.9801	0.9145	0.0656	11	0.0690
Ch+Sp @ P2	14	0.9728	0.9200	0.0528	9	0.0370
Ch+Sp @ P4	5	0.9506	0.9109	0.0397	3	0.0511
Ch+Sp @ P3+P4	14	0.9716	0.9418	0.0298	6	0.0847
Ch+Sp @ P2+P3+P4	15	0.9661	0.9291	0.0370	10	0.0440
Spatial→Channel	15	0.9622	0.9255	0.0368	9	0.0468
Kernel 3×3	18	0.9692	0.9218	0.0474	10	0.0320
Kernel 5×5	18	0.9856	0.9200	0.0656	5	0.0399
ECA	13	0.9467	0.9291	0.0176	13	0.0482
CBAM	9	0.9218	0.8855	0.0363	13	0.0869
Coordinate	5	0.9295	0.9218	0.0077	4	0.0954

Convergence. Attention did not accelerate the *start* of training: in the logged runs, the head-training accuracy after the first epoch was 0.672 for the baseline but 0.30–0.63 for the models with newly initialised gates (e.g. 0.301 for $k = 3$, 0.376 for P2+P3+P4, 0.524 for P3+P4). A plausible reason (not tested separately) is that freshly initialised sigmoid gates start near 0.5 and rescale the pretrained features. Once fine-tuning began, channel+spatial at P3 reached 90% validation accuracy earliest (epoch 3, versus epoch 6 for the baseline).

Stability and overfitting. The baseline model showed the highest stability of the validation loss (std 0.027 during the last 10 epochs); channel+spatial (0.090), Coordinate (0.095), CBAM (0.087) and P3+P4 (0.085) had higher fluctuations. Self-attention, 5×5 and P1 models showed the highest train-validation losses difference (approximately 0.064–0.066), that is, the highest level of overfitting. The negative values of the *Channel*, *Spatial* and *Combined* models are explained by the fact that the best epoch of the model is the first fine-tuning epoch when the training accuracy was evaluated with augmentation and dropout.

13. Confusion Matrix Analysis

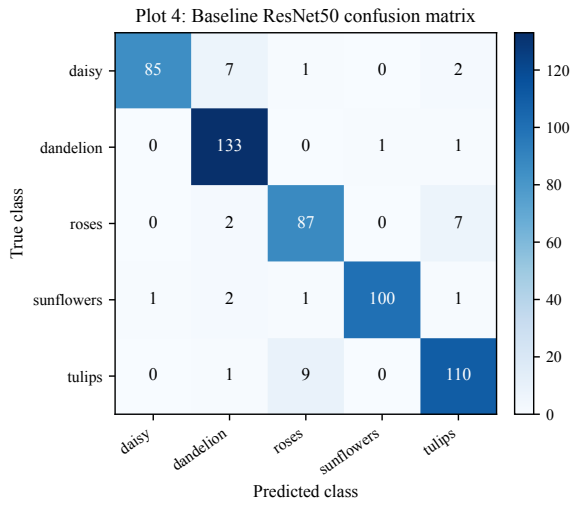

Figure 14: Plot 4: Confusion matrix of the baseline ResNet50 (test set).

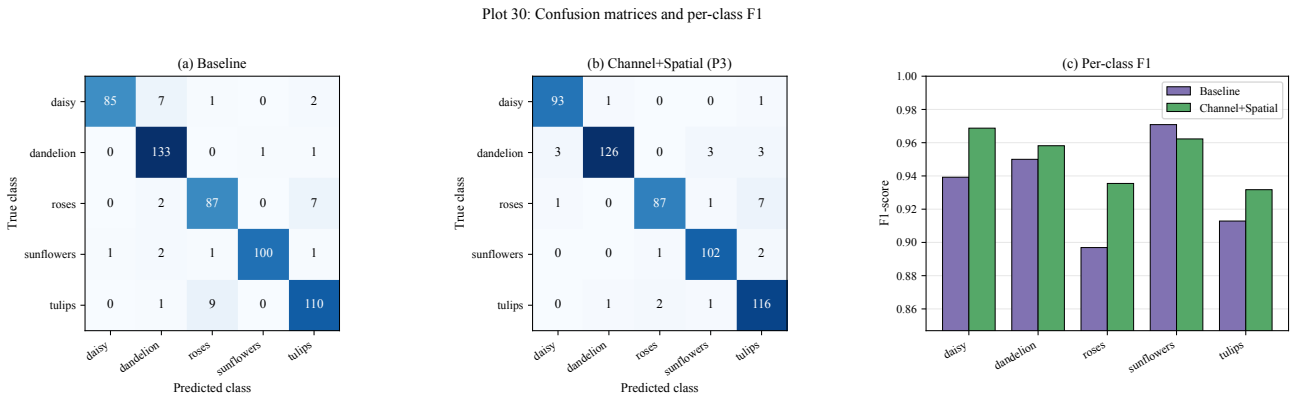

Figure 15: Plot 30: Confusion matrices of the baseline and the channel+spatial (P3) model, and per-class F1.

Table 14: Per-class recall and F1: baseline versus channel+spatial at P3.

Class	Recall (base)	Recall (Ch+Sp)	F1 (base)	F1 (Ch+Sp)	ΔF1
daisy	0.8947	0.9789	0.9392	0.9688	+0.0295
dandelion	0.9852	0.9333	0.9500	0.9582	+0.0082
roses	0.9062	0.9062	0.8969	0.9355	+0.0386
sunflowers	0.9524	0.9714	0.9709	0.9623	−0.0086
tulips	0.9167	0.9667	0.9129	0.9317	+0.0189

Observations. Sunflowers and dandelions are the most accurately classified classes, while roses and tulips are the hardest. The most common mistakes of the baseline are: tulips→roses (9 pictures), roses→tulips (7) and daisy→dandelion (7). With channel+spatial attention, tulips→roses is no longer one of the top mistakes and daisy→dandelion mistakes disappear from it (daisy recall increases from 0.895 to 0.979), while roses→tulips remains at 7. The number of total mistakes decreases from 36 to 27. The most positive influence on F1 has roses (+0.039), and the negative ones have sunflowers (-0.009) and dandelion (-0.052 due to its confusion with tulips and sunflowers, 3 images each). So, the improvement is not uniform and mainly concentrated on the roses, tulips and daisy classes.

The average improvements over all 16 attention models are as follows: only roses improve consistently (ΔF1=0.018), while sunflowers (-0.018) and dandelion (-0.014) decrease in average; daisy and tulips do not change more than 0.003.

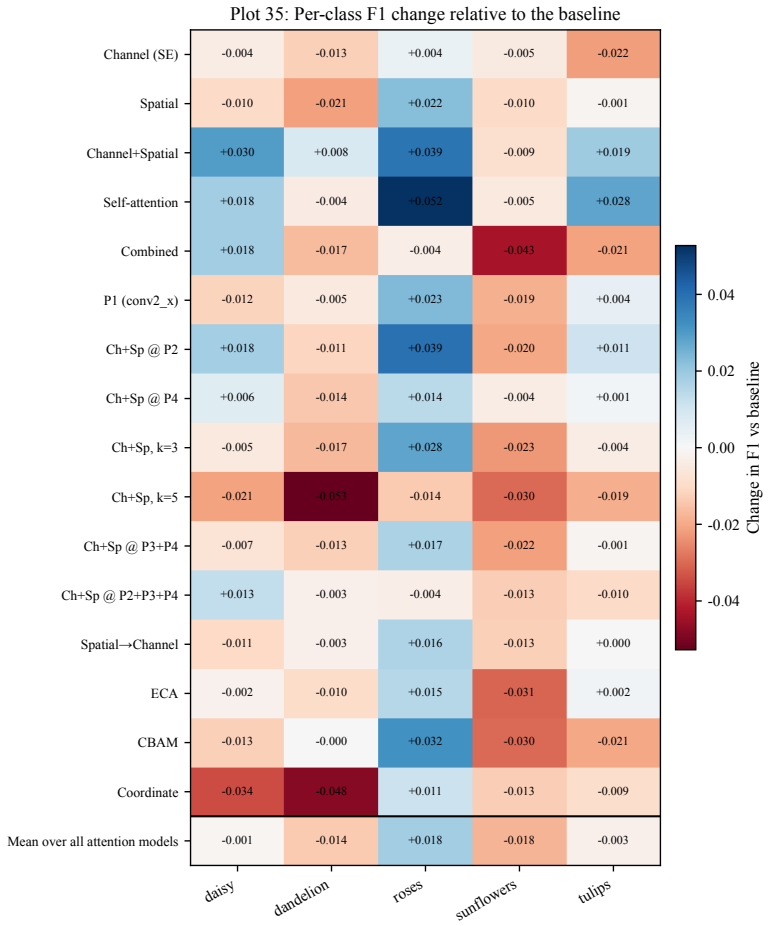

Figure 16: Plot 35: Per-class F1 change of every attention model relative to the baseline.

14. Grad-CAM Analysis

Grad-CAM was performed on the $7 \times 7 \times 2048$ feature map before global average pooling in the predicted class, and resized to 224×224 then overlayed on top of the input image. The five fixed test images used were such that three images were correctly predicted by both the baseline and the channel+spatial model (dandelion #311, tulips #322, roses #137), while two images were misclassified by the channel+spatial model (tulips #351, predicted roses by both models; dandelion #376, correct for the baseline but predicted daisy by the channel+spatial).

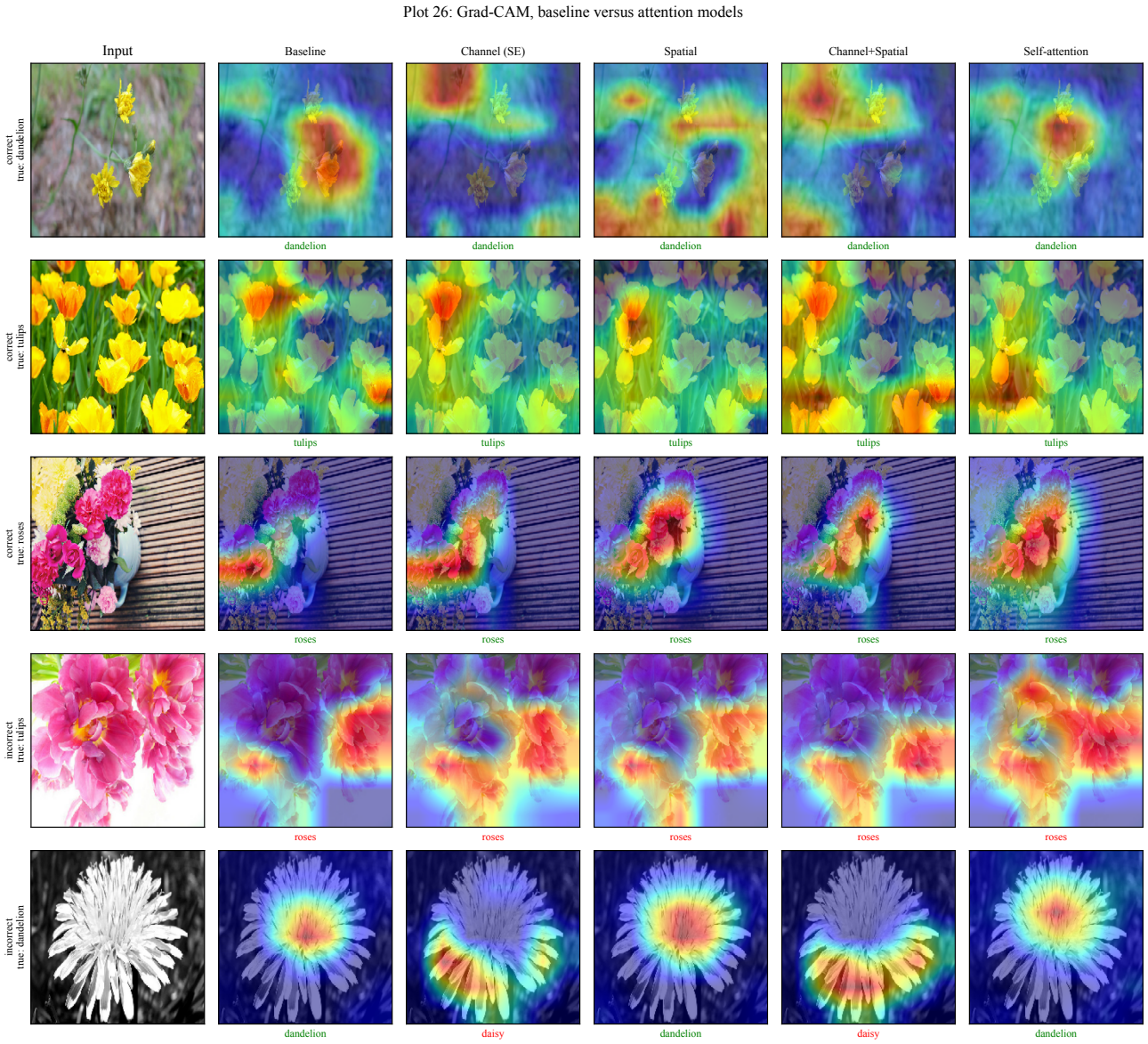

Figure 17: Plot 26: Grad-CAM for the baseline, channel, spatial, channel+spatial and self-attention models on the same five test images (green label = correct, red = wrong).

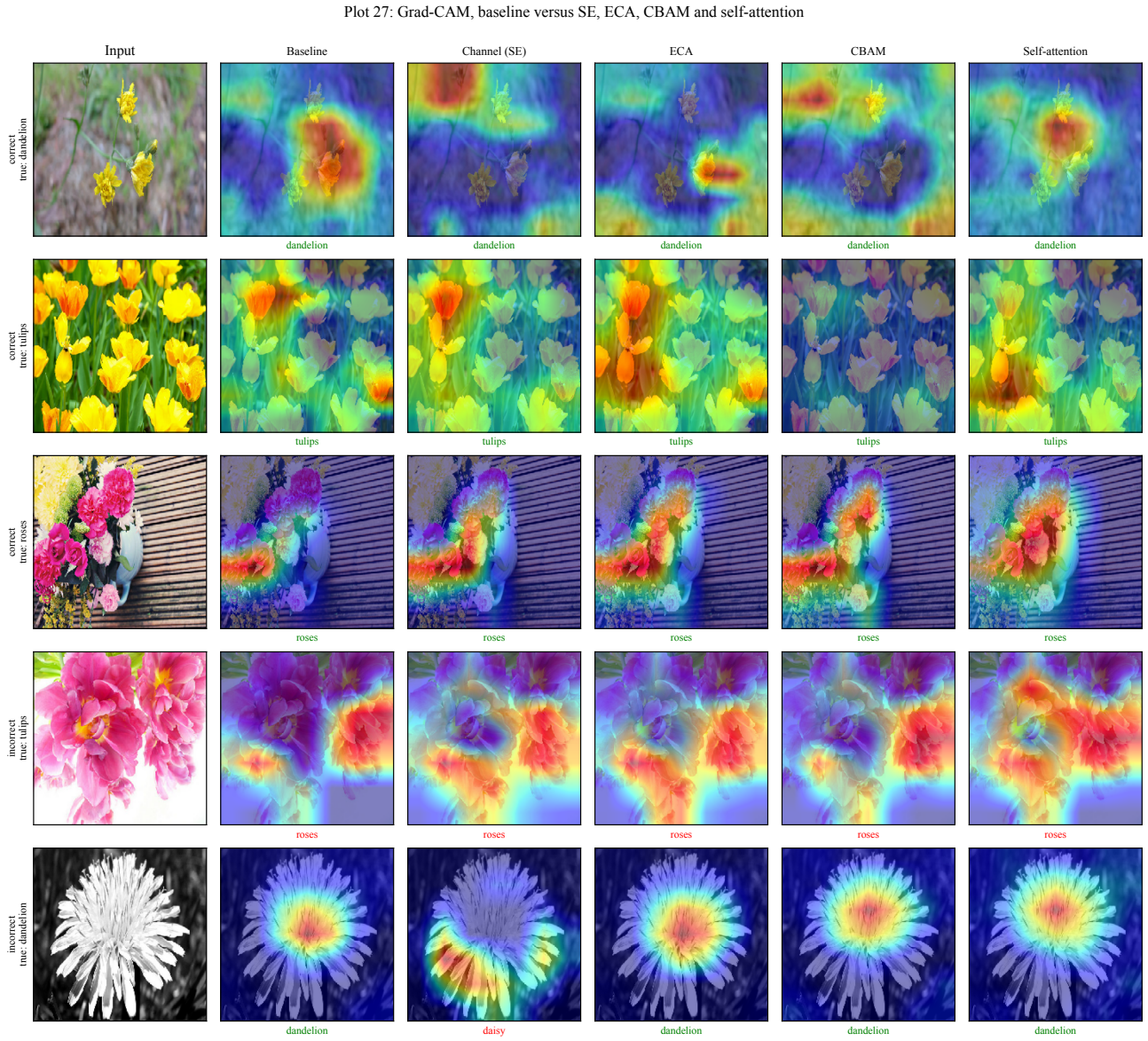

Figure 18: Plot 27: Grad-CAM for the baseline, SE, ECA, CBAM and self-attention models.

As one example set of images would not suffice, the shape of each Grad-CAM map was calculated based on *all* 551 test images. Coverage refers to the percentage of pixels with values over 0.5 on the normalised map, compactness is 1− normalised entropy (greater means more concentration) and centre ratio is the average value in the central window relative to the mean of the entire map. (tf_flowers lacks segmentation masks, thus they serve as estimates of focus, rather than a measure of whether there is a flower or not.)

Table 15: Grad-CAM statistics (correct / incorrect predictions).

Model	n (correct / incorrect)	Coverage	Compactness	Centre ratio
Baseline	515 / 36	0.173 / 0.129	0.187 / 0.215	1.804 / 1.255
Channel (SE)	510 / 41	0.195 / 0.164	0.161 / 0.204	1.805 / 1.341
Spatial	512 / 39	0.184 / 0.149	0.178 / 0.196	1.830 / 1.396
Channel+Spatial	524 / 27	0.179 / 0.156	0.178 / 0.187	1.796 / 1.115
Self-attention	524 / 27	0.214 / 0.153	0.146 / 0.198	1.700 / 1.247
ECA	512 / 39	0.177 / 0.139	0.177 / 0.235	1.844 / 1.618
CBAM	511 / 40	0.166 / 0.148	0.192 / 0.191	1.859 / 1.468
Coordinate	504 / 47	0.320 / 0.264	0.116 / 0.137	1.655 / 1.227
Ch+Sp @ P2	518 / 33	0.187 / 0.122	0.179 / 0.229	1.921 / 1.402
Ch+Sp @ P4	515 / 36	0.218 / 0.160	0.128 / 0.141	1.728 / 1.204

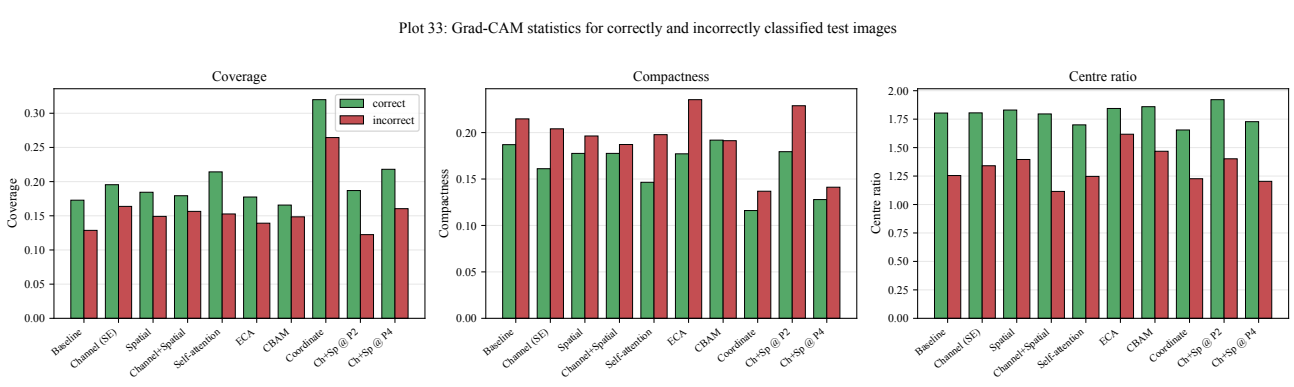

Figure 19: Plot 33: Grad-CAM statistics for correctly and incorrectly classified test images.

Interpretation.

- Grad-CAM of the channel+spatial model is very similar to the baseline's in coverage (0.179 vs. 0.173), compactness (0.178 vs. 0.187) and centre ratio (1.796 vs. 1.804): increased classification accuracy is *not* associated with more compact or more centred Grad-CAM area.
- Self-attention and P4 attention have less centred Grad-CAMs (centre ratio about 1.3 and 1.4), whereas Coordinate attention yields the least centred map (centre ratio 0.757): in all three cases evidence is distributed over the largest part of the image.
- Centre ratios are about 1.7-1.9 for *all* models including the baseline, so the centre bias is mostly caused by photographing flowers near the centre of the picture and is *not* related to attention at all.
- Grad-CAM of misclassified pictures shows lower centre ratio compared with correct classifications (baseline 1.255 vs. 1.804; channel+spatial 1.115 vs. 1.796) and lower coverage. That is, classification errors are made based on visual evidence located far from the centre of the picture. The number of such images is small (only 27-47 per model).

15. Internal Attention-Map Analysis

Internal attention maps are derived by the module itself during the forward pass, while Grad-CAM is derived after the prediction is done based on the gradients of the class score (post-hoc explanation).

15.1 Feature maps and spatial attention (channel+spatial at P3)

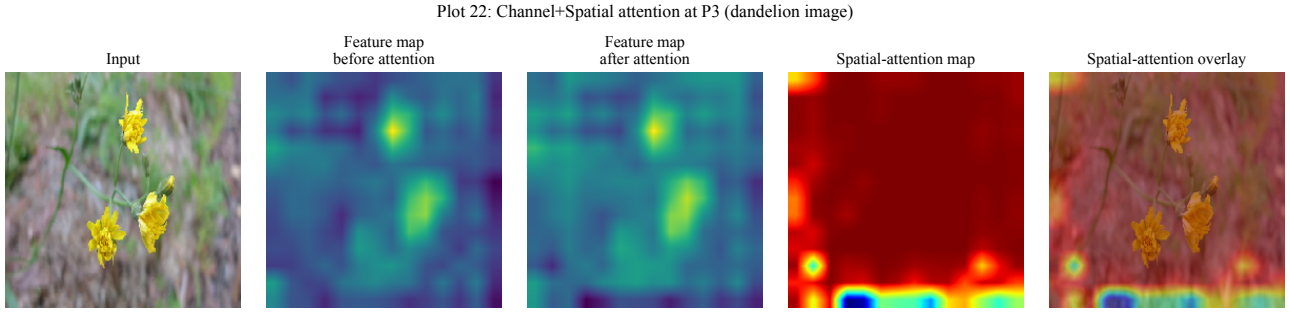

Figure 20: Plot 22: Input, feature map before attention, feature map after attention, spatial-attention map M_s and overlay.

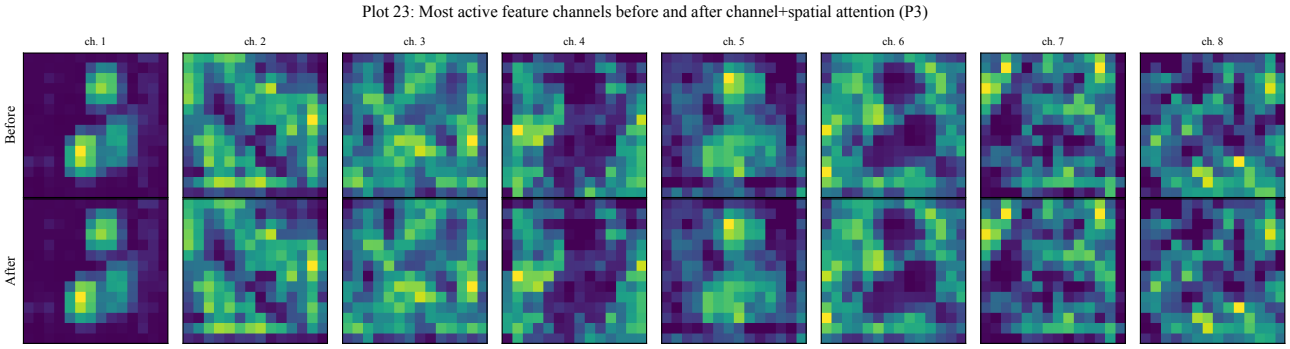

Figure 21: Plot 23: The eight most active feature channels before and after channel+spatial attention.

15.2 Channel-attention weights

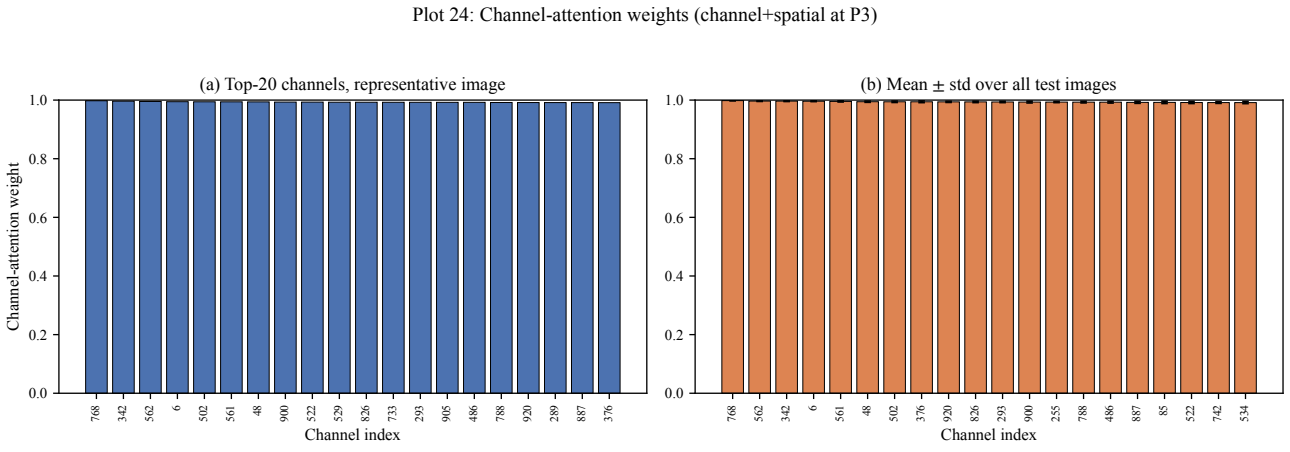

Figure 22: Plot 24: (a) top-20 channel-attention weights for a representative image; (b) mean ± std over all test images for the 20 channels with the largest mean weight.

For the channel+spatial model, weights at P3 (1024 channels) have mean 0.608, take all values between 0.000 and 1.000, and 62.7% of the weights are above 0.5. Weights change dramatically *between channels* for one image (mean per-image std 0.378), while changing very little *between images* (mean per-channel std 0.038), i.e. at P3 channel gating is nearly a fixed re-weighting of channels.

Plot 34: Distribution of channel-attention weights

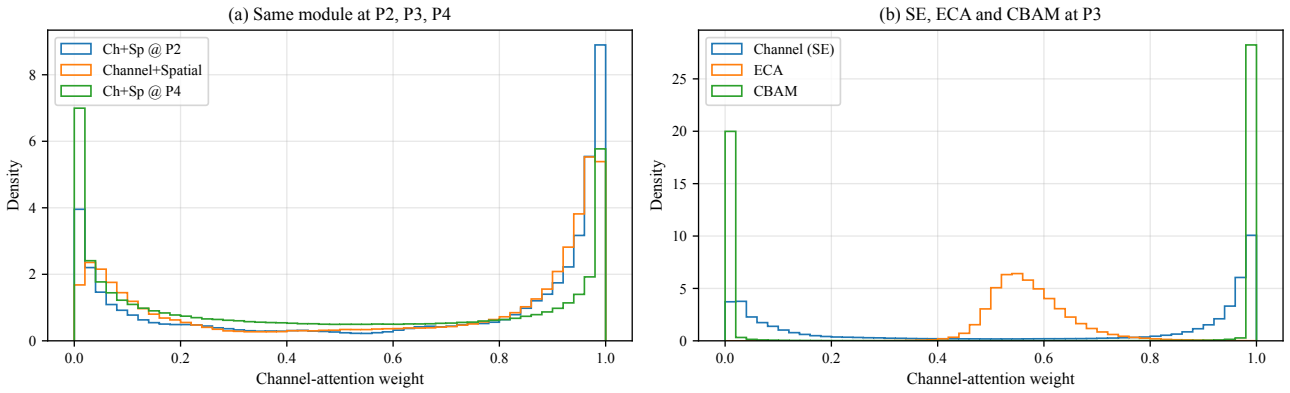

Figure 23: Plot 34: Distribution of channel-attention weights. (a) The same module at P2, P3 and P4; (b) SE, ECA and CBAM at P3.

Table 16: Channel-attention weight statistics over all test images. Selectivity = per-image std across channels; input dependence = per-channel std across images; near-neutral = fraction of weights in $[0.4, 0.6]$.

Model	Channels	Mean weight	Selectivity	Input dependence	Near-neutral	Weights > 0.9	Weights < 0.1
Ch+Sp @ P2	512	0.630	0.384	0.027	0.053	0.431	0.193
Ch+Sp @ P3	1024	0.608	0.378	0.038	0.066	0.393	0.188
Ch+Sp @ P4	2048	0.466	0.372	0.281	0.101	0.224	0.277
Channel (SE) P3	1024	0.599	0.412	0.027	0.037	0.461	0.258
ECA P3	1024	0.575	0.076	0.055	0.688	0.003	0.000
CBAM P3	1024	0.582	0.490	0.005	0.002	0.576	0.412

Most channels have high weights, but many channels are turned off (around 19-28% with weight below 0.1). This behavior varies by depth; at P2 and P3, the weights depend little on the inputs (input dependence of 0.027 and 0.038), while at P4 they become heavily image-dependent (0.281) and less than half mean (0.466). Of the pure-channel models, the ECA gate has neutral values most of the time (68.8% of its weights are between 0.4 and 0.6), while CBAM's is binary (57.6% above 0.9, 41.2% below 0.1).

15.3 Spatial-attention maps and the effect of kernel and order

Table 17: Statistics of the spatial-attention map M_s over all test images. Gate range = max − min of the raw map (low = nearly constant gate).

Study	Model	Mean gate	Gate range	Coverage	Compactness	Centre ratio
Kernel	$k = 3$	0.900	0.718	0.925	0.008	1.080
Kernel	$k = 5$	0.983	0.439	0.973	0.003	1.041
Kernel	$k = 7$	0.987	0.271	0.974	0.003	1.052
Order	Channel→Spatial	0.987	0.271	0.974	0.003	1.052
Order	Spatial→Channel	0.986	0.401	0.978	0.003	1.038
Position	P2 (28×28)	0.981	0.726	0.981	0.001	1.026
Position	P3 (14×14)	0.987	0.271	0.974	0.003	1.052
Position	P4 (7×7)	0.998	0.058	0.968	0.007	1.044

Figure 24: Plot 32: Spatial-attention maps at P3 for $k = 3, 5, 7$ and for the Spatial→Channel order.

The spatial gate is near saturation, since its average value is 0.90–0.998, with 92-98 percent of the locations having values greater than 50percent of the normalised range, so the gate rescales the features rather than eliminating portions of it. The wider kernel results in a smoother and more constrained gate range (0.718, 0.439, 0.271 for kernels 3x3, 5x5 and 7x7). The 3x3 kernel produces the most selective gate range, but not the highest accuracy, so selective gates do not imply better classifier performance.

15.4 Self-attention response

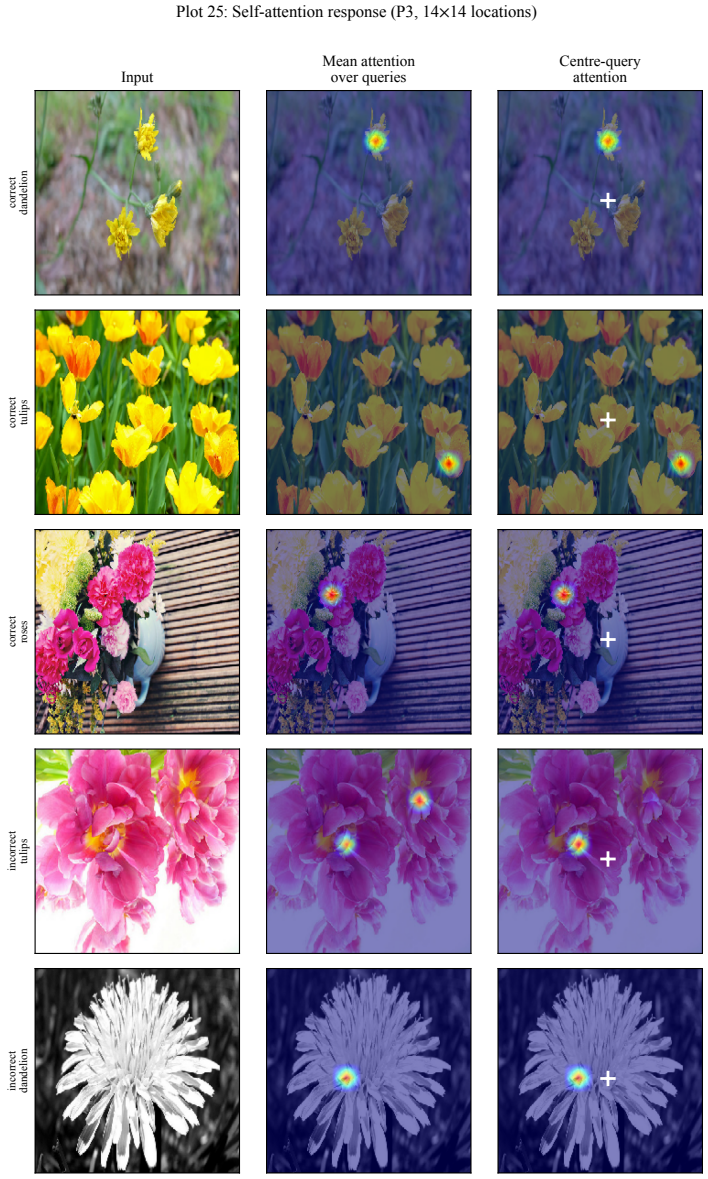

Figure 25: Plot 25: Self-attention response at P3 ($14 \times 14 = 196$ locations): attention received (mean over all queries) and the attention of the centre query (white +).

The learned scale of the residual branch was at $\gamma = -0.0588$, which is a low value, meaning that the attention branch only applies a small correction on top of the pretrained feature vectors. The mean attention maps show the locations attended by all other locations, while the center-query maps show distant interactions with a single location.

15.5 Attention at P2, P3 and P4

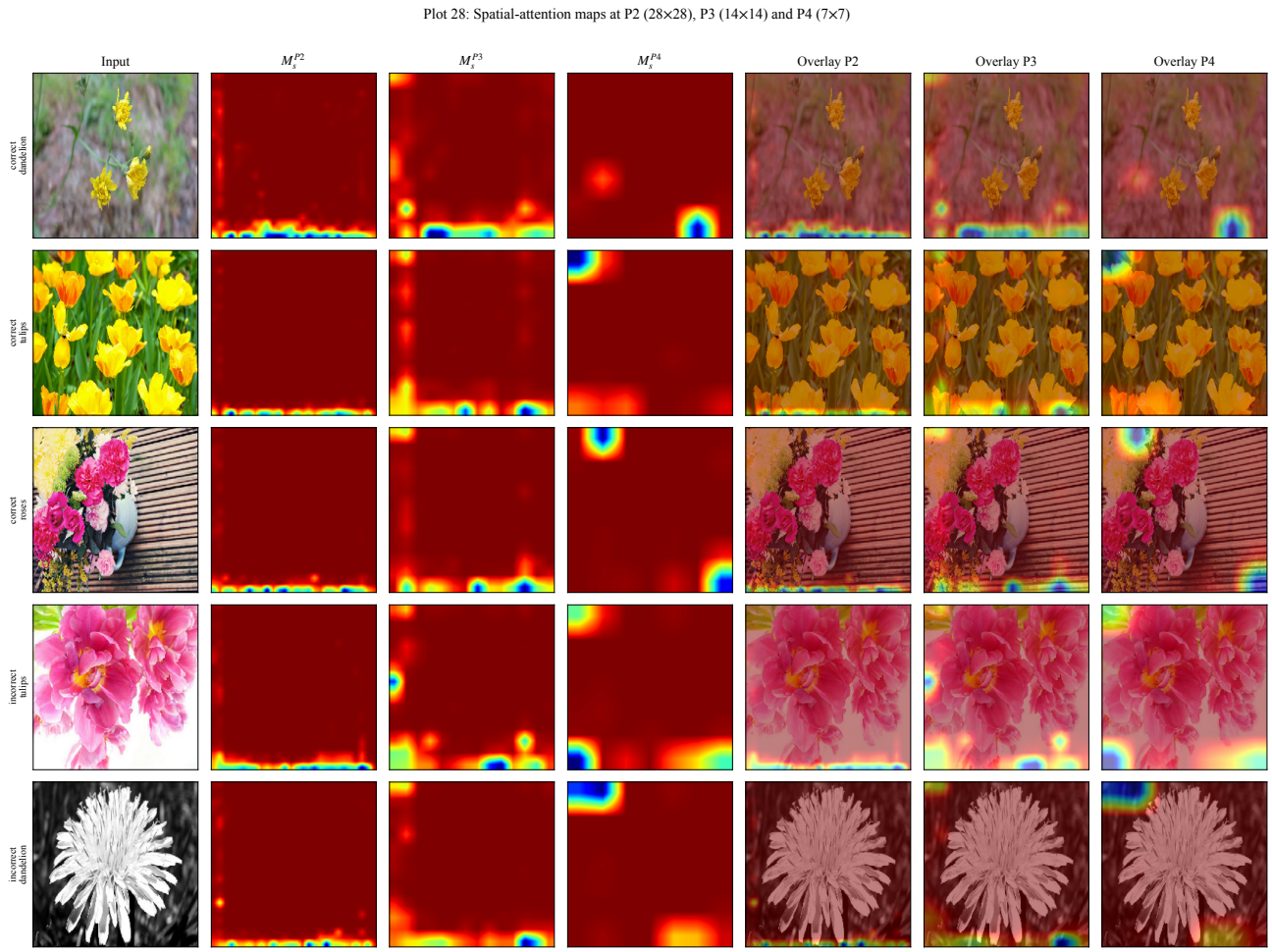

Figure 26: Plot 28: Spatial-attention maps M_s^{P2}, M_s^{P3}, M_s^{P4} and overlays on the same five test images.

The statistics in Table 17 illustrate how the maps vary depending on the depth: for the gate range, the value decreases from 0.726 in P2 to 0.271 in P3 and 0.058 in P4, and the mean gate increases toward 1 (0.981, 0.987, 0.998). In P2, the map is more structured (28×28 locations), while in P4, the 7×7 gate is almost constant and thus alters the features slightly; this is in line with the P4 model performing similarly to the baseline (0.9347).

15.6 Internal attention versus Grad-CAM

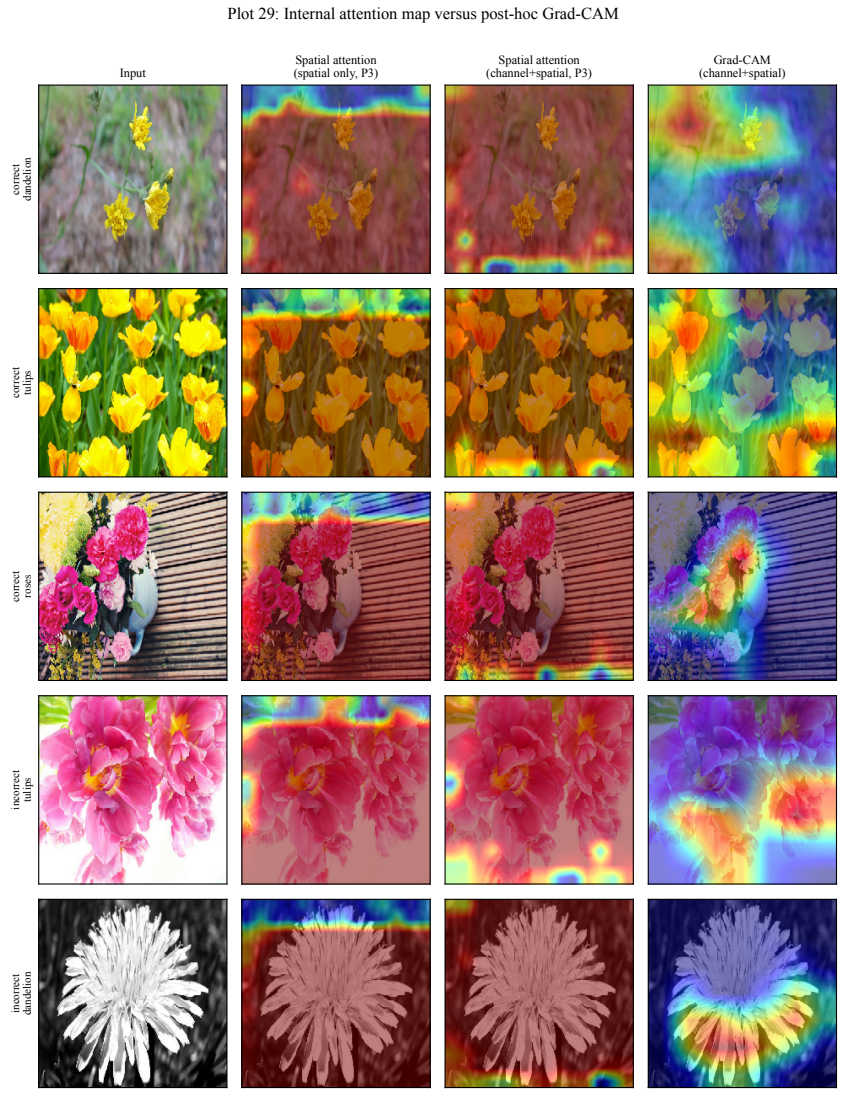

Figure 27: Plot 29: Internal spatial-attention maps (spatial and channel+spatial models) versus the post-hoc Grad-CAM of the channel+spatial model.

The two types of heatmaps can be generated in different ways, so that agreement between the two is not required: the internal heatmap is nearly saturated and is calculated based on statistics of channels aggregated in 14×14, whereas Grad-CAM is class-dependent and is calculated in the final stage of the network (7×7), being very sparse (coverage ~ 0.18).

16. Qualitative Comparison Across Attention Mechanisms

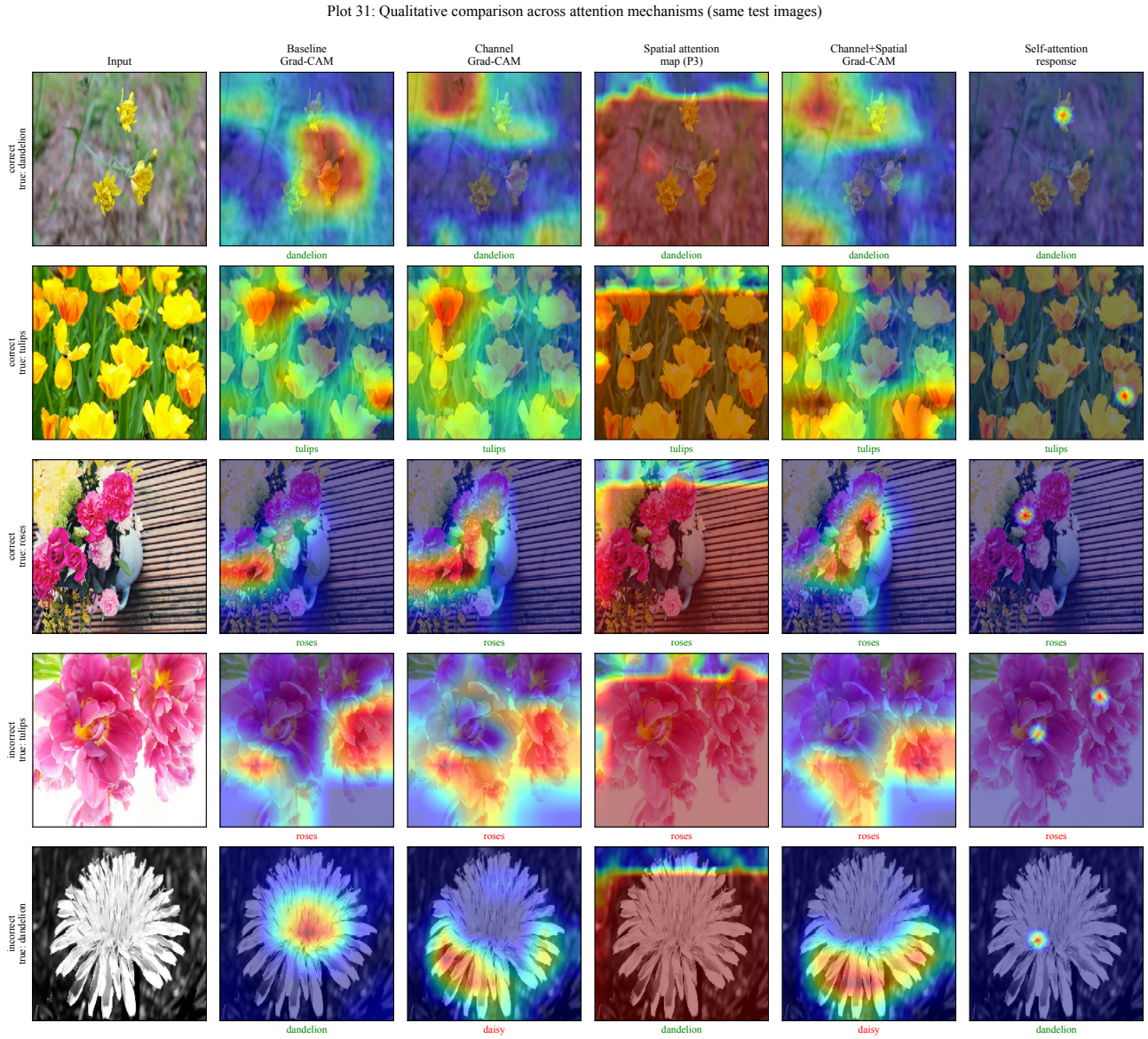

Figure 28: Plot 31: Input, baseline Grad-CAM, channel Grad-CAM, spatial-attention map, channel+spatial Grad-CAM and self-attention response for three correctly and two incorrectly classified test images.

This comparison is based on foreground localization, background suppression, activation compactness and coverage, as well as correct versus incorrect images. The numerical aspects of these parameters are described in Table 11 (Grad-CAM statistics): compactness and coverage are essentially the same for the baseline, SE, spatial and channel + spatial models, and two images that have been misclassified by channel + spatial (tulips classified as roses, dandelion classified as daisy) happen to belong to those with low centre ratio. Image #376 with dandelion proves that attention may make even a correct baseline classification incorrect, which is why such examples should be considered alongside test-set statistics.

17. Additional Attention-Mechanism Comparison

ECA, CBAM and Coordinate Attention were compared with the mechanisms above at P3, using the same backbone and protocol. The “Channel / SE” row is the SE-style module A1.

Table 18: Comparison of attention mechanisms at P3.

Attention	Accuracy	Precision	Recall	Macro F1	Added params	Total params	ms/img
No attention	0.9347	0.9385	0.9310	0.9340	0	23,597,957	8.154
Channel / SE	0.9256	0.9283	0.9243	0.9260	132,160	23,730,117	7.708
Spatial	0.9292	0.9335	0.9278	0.9298	99	23,598,056	8.597
ECA	0.9292	0.9293	0.9321	0.9288	5	23,597,962	7.911
CBAM	0.9274	0.9397	0.9215	0.9277	132,259	23,730,216	8.262
Self-attention	**0.9510**	**0.9534**	**0.9509**	**0.9518**	1,310,721	24,908,678	7.551
Coordinate	0.9147	0.9229	0.9109	0.9153	100,512	23,698,469	7.998

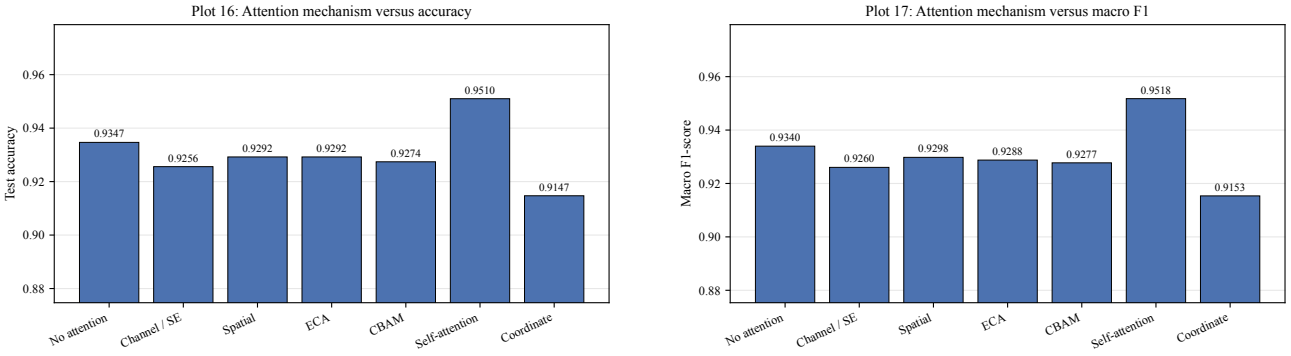

Figure 29: Plots 16 and 17: Attention mechanism versus accuracy and macro F1.

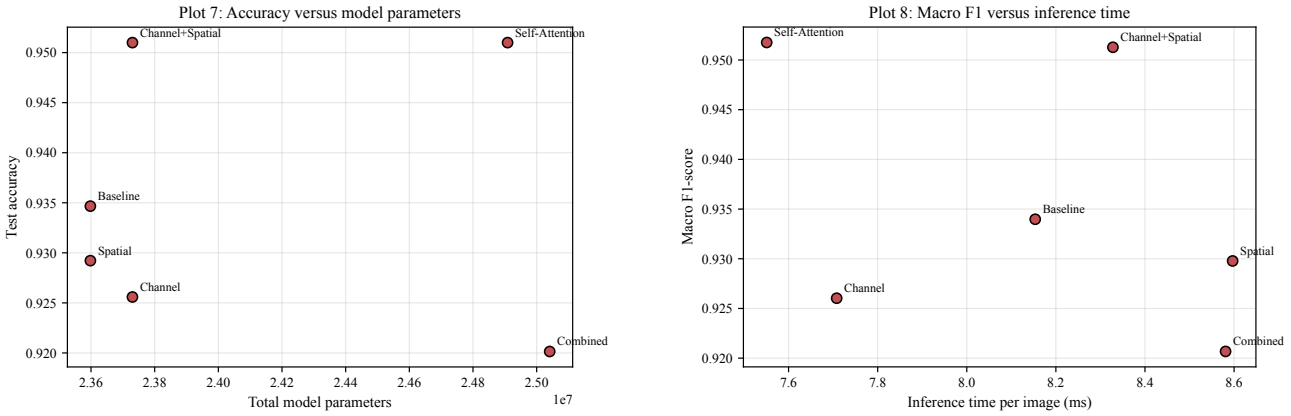

Figure 30: Plots 7 and 8: Accuracy versus parameters and macro F1 versus inference time for the attention-type study.

Of the added mechanisms, self-attention was the best, but the already existing lightweight modules (SE 0.9256, ECA 0.9292, CBAM 0.9274, Coordinate 0.9147) all performed either on par or worse than the baseline (0.9347). The coordinate attention mechanism performed the worst, and it had the most diffuse Grad-CAM visualization. Heavier module did not mean that it would perform better, and lightweight module also did not mean that it would be comparable to the baseline.

18. Discussion

Is the improvement real? The test set contains 551 images, so one image is worth 0.18 percentage points. Each model was trained once. A paired bootstrap (2,000 resamples) and an exact McNemar test against the baseline give the following picture.

Table 19: Accuracy change versus the baseline: paired bootstrap 95% interval and McNemar test (fixed = wrong→right, broken = right→wrong).

Model	Δ acc.	95% CI low	95% CI high	Fixed / broken	McNemar p	CI excludes 0
Channel (SE)	−0.0091	−0.0290	+0.0109	13 / 18	0.473	no
Spatial	−0.0054	−0.0236	+0.0145	13 / 16	0.711	no
Channel+Spatial P3	+0.0163	−0.0036	+0.0381	21 / 12	0.163	no
Self-attention	+0.0163	−0.0036	+0.0363	21 / 12	0.163	no
Combined	−0.0145	−0.0345	+0.0073	13 / 21	0.229	no
Ch+Sp @ P1	−0.0018	−0.0236	+0.0200	17 / 18	1.000	no
Ch+Sp @ P2	+0.0054	−0.0127	+0.0254	16 / 13	0.711	no
Ch+Sp @ P4	+0.0000	−0.0200	+0.0200	16 / 16	1.000	no
Ch+Sp, $k = 3$	−0.0054	−0.0236	+0.0127	12 / 15	0.701	no
Ch+Sp, $k = 5$	−0.0290	−0.0508	−0.0073	10 / 26	0.011	**yes**
Ch+Sp @ P3+P4	−0.0054	−0.0272	+0.0163	17 / 20	0.743	no
Ch+Sp @ P2+P3+P4	−0.0036	−0.0236	+0.0163	16 / 18	0.864	no
Spatial→Channel	−0.0018	−0.0236	+0.0200	18 / 19	1.000	no
ECA	−0.0054	−0.0254	+0.0163	15 / 18	0.728	no
CBAM	−0.0073	−0.0254	+0.0091	10 / 14	0.541	no
Coordinate	−0.0200	−0.0417	+0.0018	12 / 23	0.090	no

Neither the best model configurations (channel+spatial and self-attention at P3, +1.63 points) nor the worst one (self-attention at P5, −1.19 points) are significantly better/worse than the baseline (all intervals contain zero), and the only significant difference between models is a deficit (5×5 kernel). Thus, these results support the idea that intermediate features are the most valuable level for applying the attention mechanism, but they do not prove it conclusively (repeated-seeds experiment is required). In addition, only test-set sampling noise is taken into account, and training run-to-run variance is not, which, in all likelihood, explains the non-monotonic kernel effect.

Why does attention work at P3 but not at other points? At P3 the features are semantically rich enough for effective channel re-weighting and at the same time are spatially resolved enough to apply spatial gating. At P1 the features are low-level and the gate does not have enough useful information to learn. At P4 the spatial gate (size 7×7) is nearly constant (0.058 range) and the head already has global pooling, so attention adds parameters (526,563) but no benefits.

Can increased attention improve performance? No, the combination of attention layers (P3+P4, P2+P3+P4) and three-layer attention module caused decrease in accuracy compared to one P3 attention layer but with an increase of up to 1.4M parameters.

Are quantitative and qualitative results consistent? Partially, the accuracy improvement of the channel and spatial attention modules is limited to roses, tulips, and daisy (Section 13). On the other hand, the Grad-CAM statistics for channel+spatial do not change from the baseline (Section 14), therefore this improvement cannot be credited to better visualization. Spatial gates are almost saturated, hence little contribution to background suppression through gating mechanism.

Limitations. One training per model, a small test dataset, one shared GPU for benchmarking and no segmentation masks to evaluate foreground focus.

19. Questions

1. **What is the role of residual connections in ResNet50?** They sum up the block input and output ($y = F(x) + x$); this allows backpropagation through very deep networks without problems and enables a block to learn just a little correction.
2. **How is `conv2_x` different from `conv5_x`?** `conv2_x` is the first stage (56×56, 256 channels) and computes low-level features like edges and colours; `conv5_x` is the last stage (7×7, 2048 channels) and learns high-level, class-related semantics.

3. **Why is there resolution degradation with depth?** Each stage contains strided convolutions that halve the resolution; this increases receptive field size, adds some translational invariance and kepdf computations feasible.
4. **Why does the number of channels grow with depth?** The resolution is degraded, so we need more channels to represent more complex patterns.
5. **What is meant by low-, intermediate- and high-level features?** Low-level: edges, colours, textures. Intermediate: repeated patterns, petal arrangement, local shapes. High-level: full object shape and class-related semantics.
6. **What is the aim of an attention mechanism?** The goal is for the network to highlight important channels or positions (locations) or their relation, and suppress others, by predicting weights for rescaling the features.
7. **What is learned in channel attention?** Which feature channels are important, based on a pooled global representation.
8. **What is learned in spatial attention?** Which spatial positions are important, based on channel-wise mean and max maps.
9. **How does channel attention differ from spatial attention?** Channel attention learns one weight per channel ($1 \times 1 \times C$), and it doesn't care about position information; spatial attention learns one weight per position ($H \times W \times 1$) for all the channels.
10. **Why can they be applied sequentially?** They complement each other ("what" first then "where") and both keep the structure of the feature tensor, such that the output of one serves as a valid input to the other. Our results show that Channel→Spatial (0.9510) was better than the two modules individually (0.9256 and 0.9292) at P3.
11. **What could make the attention mechanism operate in a way that differs between `conv3_x` and `conv5_x`?** The channels represent different information: edge/detection filters at conv3_x and semantic part detectors at conv5_x. Proved: at P2 the weights depend very little on the input image (input dependence 0.027), but at P4 – a lot (0.281), and the mean weight is 0.630 versus 0.466.
12. **Why can spatial attention be useful at an intermediate layer?** In such layer, the features have still decent spatial resolution (with dimension of 14×14 or 28×28) and include information about parts of the image; our best place for the spatial module was P3.
13. **Why is the self-attention costly for high resolution images?** At each location one needs to attend to all others, thus the attention map contains N^2 values, with $N = HW$.
14. **How does the size of the self-attention matrix change with resolution?** The size increases quadratic with the number of locations: it contains 2,401 entries per image at P4, 38,416 at P3, 614,656 at P2 and 9,834,496 at P1. That is why self-attention was not applied at P1.
15. **What effect does the insertion point have on the performance?** For channel→sapatial attention: P1 0.9328, P2 0.9401, P3 0.9510, P4 0.9347 (inverted-U shape with max at P3; the differences are within test noise).
16. **What is different in P2 attention map compared to P4 attention map?** P2 map is 28×28 and has a much wider range of values (0.726) while the P4 map is 7×7 and almost constant (range 0.058). See Plot 28.
17. **Does multi-level attention always lead to improved performance?** No. P3+P4 (0.9292) and P2+P3+P4 (0.9310) are worse than P3 (0.9510) and increased number of parameters by 659k and 692k respectively.
18. **Is Channel→Spatial attention different from Spatial→Channel?** Slightly: accuracy 0.9510 vs. 0.9328 (10 images, non-significant difference) and the spatial gate range 0.271 vs. 0.401.
19. **What effect does spatial kernel size have on the attention map?** Bigger kernels lead to smoother attention maps with smaller ranges of gates (gate ranges 0.718, 0.439, 0.271 for $k = 3, 5, 7$). The accuracy does not follow the same pattern (0.9292, 0.9056, 0.9510).
20. **SE vs ECA - What is the difference?** SE consists of two FC layers with a reduction ratio (132,160 parameters at P3), while ECA contains one 1-D convolution of adjacent channels without reducing dimensions (5 parameters).
21. **How does CBAM combine channel and spatial attention?** Sequentially - channel attention branch (MLP with an average- and max-pooled representation, sum of both) followed by the calculation of the spatial attention branch (7×7 convolution on top of the results of the channel

attention).

22. **What is the difference between self-attention and channel attention?** Channel attention calculates the global weights of each channel, while self-attention calculates pairwise relations and mixes the values.
23. **Does the module with the most parameters have the highest accuracy?** No. The combined module (1,442,980 parameters) is the least accurate of the A-series (accuracy 0.9201), while channel+spatial (132,259) equals self-attention (1,310,721) at 0.9510.
24. **How does an internal attention map differ from Grad-CAM?** The internal map is a part of the network and is calculated during the forward pass; Grad-CAM is class-specific and calculated after prediction from gradients.
25. **Why is Grad-CAM post-hoc?** It is calculated using an already trained and evaluated model without any modifications of the model itself or its predictions.
26. **Does the attention model filter out the background?** Not noticeably here. The spatial gating functions are almost fully activated (mean 0.987, coverage 0.974), and the Grad-CAM compactness and centre ratio of channel + spatial gates are the same as the baseline. (No masks are available for accurate measurement.)
27. **Do the maps of correct and incorrect predictions differ?** Yes, on average: for all ten tested models the Grad-CAM of incorrect predictions has a lower centre ratio and lower coverage (e.g. baseline centre ratio 1.255 vs. 1.804), based on 27–47 incorrect images per model.
28. **Which classes benefit most?** Roses (largest F1 gain for 13 out of 16 attention mechanisms; +0.039 for channel+spatial), followed by daisy and tulips; sunflowers and dandelion lost on average.
29. **What does a confusion matrix reveal beyond accuracy?** Classes that get mixed up with one another (e.g., tulips↔roses), and the distribution of gain among classes.
30. **How does macro F1 help beyond just accuracy?** Macro F1 treats each of the five classes equally and balances precision with recall, thereby highlighting class-specific problems.
31. **What is the accuracy-complexity trade-off?** Whether additional parameters yield any additional accuracy: channel+spatial P3 offers 0.9510 for +0.56% more parameters, but self-attention yields the same for +5.55% more parameters.
32. **What is the F1-inference time trade-off?** Whether spending more computational effort per image improves F1. All models in the study require 7.55–8.60 ms per image without correlation with F1 scores, hence no trade-off in this case.
33. **Why must every ablation be done on the same data split?** Accuracy differences otherwise could be a result of easier or harder test images.
34. **Why must the same training procedure be followed?** The optimiser, learning rate, epoch, augmentation, and fine-tuning all affect accuracy, so keeping those constant means the independent variable is the one thing that varies.
35. **What can we conclude about the comparison of Grad-CAM?** The Grad-CAM statistics for the channel+spatial model are very similar to the baseline; self-attention, P4 and coordinate attention distribute evidence across a larger region; and incorrect predictions are linked to non-centred Grad-CAM regions.

20. Conclusion

Models: 1 (Baseline), and 16 with attention (total 17 models) were trained and compared using a single protocol with a single 70/15/15 data split of tf_flowers with a pretrained ResNet50 network. Baseline achieved 0.9347 test accuracy. Channel to spatial attention and self-attention after `conv4_x` (P3) reached 0.9510 (+1.63 accuracy points); channel+spatial attention reached this value with only 132,259 additional parameters (0.56%), that is, 1/10 of the amount of self-attention. Position of attention was more important than its type: the same attention module achieved its maximum performance at P3 (P1 0.9328, P2 0.9401, P3 0.9510, P4 0.9347). Multi-stage attention, combined module, inverted attention, attention kernels of size 3×3 and 5×5, as well as known SE, ECA, CBAM and Coordinate modules did not surpass the baseline. Inference time was between 7.55 and 8.60 ms per image for all models.Channel-based attention provides consistent improvement only with the rose dataset. None of the gains made compared to the baseline is statistically significant with a single run per model using 551 test images,

and the statistics of Grad-CAM for the most successful attention model is very similar to the baseline, therefore proving that "P3 is the most promising position" is a hypothesis rather than a proven fact.

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